



GE VERNOVA

ADMS

16.25.1

Model Manager Installation Guide

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Contents

1. Introduction	5
2. Installing the Model Manager	6
2.1. Prerequisites	6
2.1.1. Uninstalling the Prior Version (Tools Kit)	6
2.2. Installation	7
2.3. Model Manager Configuration Editor and File	8
2.3.1. Model Manager Configuration File	8
2.3.2. Launching the Model Manager Configuration Editor	8
2.4. Model Manager Standalone Application Mode	10
2.4.1. Configuring the Windows Shortcut	10
2.4.2. Verifying a Standalone Installation	11
2.5. Model Manager Service Mode	13
2.5.1. Windows Shortcuts	13
2.5.2. Installing the Service Provider	14
2.5.2.1. Verifying Installation	14
2.5.2.2. Reviewing Model Manager's Event Log	17
2.5.3. Delegating Permissions for Non-Admin User to Manage Windows Service	18
2.5.4. Starting the Client and Verifying the Setup	18
2.5.5. Verifying Startup and Other User Access	19
2.5.6. Verifying Model Manager Processing	20
3. Model Manager Web Service	21
3.1. Accepting SOAP Messages from GISA	21
3.1.1. Generating and Configuring a Self-Signed Certificate	21
3.1.1.1. Creating a Self-Signed Certificate	22
3.1.1.2. Configuring a Self-Signed Certificate	22
3.1.2. Adding SSL Binding	27
3.1.3. Importing a GISA Certificate into a Trusted Root CA	28
3.1.4. Adding a Namespace Reservation for the Port	33
3.2. Sending SOAP Messages to GISA	33
3.2.1. Exporting an MMA Server Certificate for GISA Client Usage (Export without Private Keys)	34
3.2.2. Exporting the MMA Server Certificate for Future Use (Export with Private Keys)	39
3.3. SFTP File Transfer from GISA to Model Manager	45
3.3.1. Installing VanDyke SecureFX	46
3.3.1.1. Generating Keys using VanDyke SecureFX	47
3.3.2. Installing VanDyke VShell	51
3.3.2.1. Configuring VShell Virtual Roots (Defining SFTP Folders)	51
3.3.3. Configuring the VShell Authentication	53
3.3.4. Starting the VShell Service	54
3.3.5. Validating the SSH Connection	55

4. Configuring HTTPS for the REST Web Service	58
4.1. Creating Certificates	58
4.2. Accepting REST Messages from External Systems	59
4.2.1. Configuring MMA Server Certificate	59
4.2.2. Adding SSL Binding	64
4.2.3. Importing External System CA Certificate to Trusted Root CA Folder	65
4.3. Sending REST messages to External Systems	70
4.3.1. Exporting MMA Server Certificate for Client Authentication	70

1. Introduction

The Model Manager tool performs "Model Management" for the ADMS system. The Model Manager creates station model files from collections of input files by invoking model "compilers" (Converter and Navigation Overview Builder) to create binary model files that define the distribution system base model. The resulting station model files are then stored in the appropriate folder for further review or transfer into the online ADMS servers.

In an effort to accommodate varied customer needs, the Model Manager can operate in two modes: as a Windows service to constantly scan for new files, or as a standalone application to be run on demand. The ADMS tools installation kit installs four programs: the Model Manager Service, the Model Manager Service Client, the Model Manager Application, and the Model Manager Configuration Editor, along with start menu shortcuts to support the different operating modes.

Model Manager Service

The Model Manager Service runs as a Windows service. It contains the business logic (backend) of the Model Manager. The business logic and software implementation are shared between the Model Manager Service and the Model Manager Application (standalone program).

The Model Manager Service Client is used with the Model Manager Service to view and control the Model Manager processing.

Model Manager Service Client

The Model Manager Service Client provides the Model Manager user interface when operating in Windows service mode. It connects to the Model Manager Service using underlying Windows communication services.

Model Manager Configuration Editor

The Model Manager Configuration Editor is used to configure the Model Manager.

Model Manager Application

The Model Manager application provides the combined functionality of the Model Manager Service and Service Client. The Model Manager Application is for those customers who want to operate the Model Manager as an interactive program and do not require the benefits of running Model Manager as a service (i.e., user independence, automatic startup, etc.).

The Configuration Editor interface is available to the standalone client via the corresponding menu option.

2. Installing the Model Manager

The Model Manager software is packaged in the ADMS Tools installation kit. It can be configured to operate in one of the following modes:

- Standalone application
- Windows service provider and client applications (recommended)

While these two configurations can be installed at the same time, they cannot run at the same time. In addition, the Windows service installation allows for 24x7 availability, and is the recommended approach.

The installation kit provides shortcuts for both configurations. To avoid confusion, GE recommends removing the shortcuts you do not plan to use after choosing your preferred method.

Installation and setup of both the standalone application and service provider are described in this chapter. Both installations share the same configuration editor and configuration file.

For more information about how to operate the Model Manager, refer to the *Model Manager User Guide*.

2.1. Prerequisites

The following prerequisites apply to all supported and certified configurations of the Model Manager. For a list of the platforms, refer to the "Supported Configurations" chapter in the *ADMS Release Notes*.

To make sure you have the latest information, including whether each prerequisite is for the Model Manager, review the "Certified Third-Party Products and Packages" section in the *ADMS Release Notes*.

To install the Model Manager, no unique prerequisites exist that are not already common to other ADMS tools.

Caution

Prior to installing an upgrade, you may want to preserve the previous version's configuration file `ModelManager.xml`.

2.1.1. Uninstalling the Prior Version (Tools Kit)

Prior to upgrading your Model Manager software (Tools kit) you should ensure the Model Manager service is stopped if you have chosen to run Model Manager as a service.

To stop the Model Manager service:

1. Run `services.msc` from the Windows Start menu or command line.

The Service Management Console opens.

2. Stop the ADMS Model Management Service service.

Or:

- From the Windows Start menu, click **Eterra> Distribution> Tools> Modeling Tools> Stop Model Manager Service**.

Uninstall the currently installed Tools kit:

1. From the Windows Start menu, click Control Panel, and then open Programs and Features.
2. Uninstall ADMS Tools.

2.2. Installation

Run the ADMS_tools_X64.msi installer. During installation you have the option to only install the Model Manager.

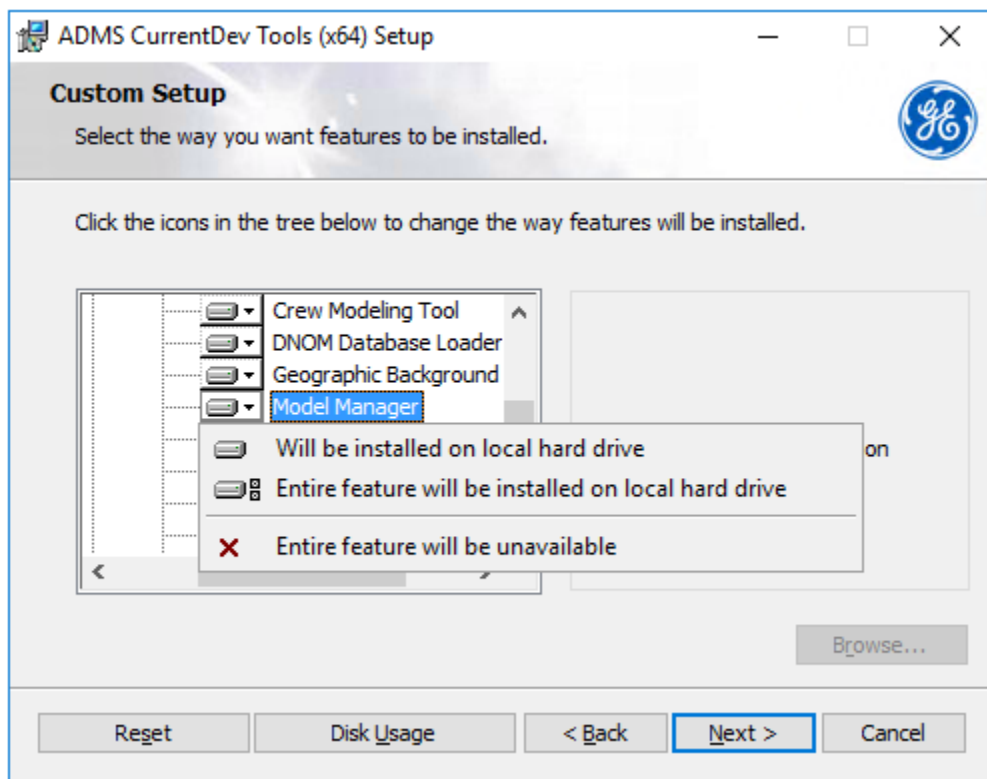


Figure 1. Installing the Model Manager

If you are not using the default expansion pack configuration, which includes the setting of the %IDMS_ROOT% environment variable, you should edit the configuration using the Configuration Editor, as described in [Model Manager Configuration Editor and File](#).

Verify the installation using the standalone application mode via the Model Manager Application

shortcut, as described in [Model Manager Standalone Application Mode](#).

Once the standalone application is verified, continue with the configuration and verification of the Windows Service provider and client mode, as described in [Starting the Client and Verifying the Setup](#).

To minimize confusion, delete the Windows shortcut to the standalone `ModelManagerApplication.exe`.

Note

The `IDMS_ROOT` environment variable must be system scoped (added to the system user account) in order for the Model Manager Service to run properly.

2.3. Model Manager Configuration Editor and File

The Model Manager Configuration Editor is installed with the tools kit. Once the system is configured, the Configuration Editor can be deleted (optionally) from end-user profiles.

2.3.1. Model Manager Configuration File

The configuration file, `ModelManager.xml`, is installed by default in the following location:

`%IDMS_ROOT%\ToolsWorkspace\ModelManagerFolders`

All edits to the configuration file are made via the Configuration Editor.

As a short term workaround, edit and verify that all the paths in the XML are resolvable and reachable by the `NetworkService` account.

For example, if `%IDMS_ROOT%` is in the path, it needs to be a system-scoped environment variable. Otherwise, replace `%IDMS_ROOT%` in the configuration file with corresponding absolute paths.

2.3.2. Launching the Model Manager Configuration Editor

From the Windows Start menu, navigate to **Eterra> Distribution> Tools> Model Manager Configuration Editor**.

If the configuration file is not already provided in the shortcut's Target field, you may need to browse for and open the configuration file manually.

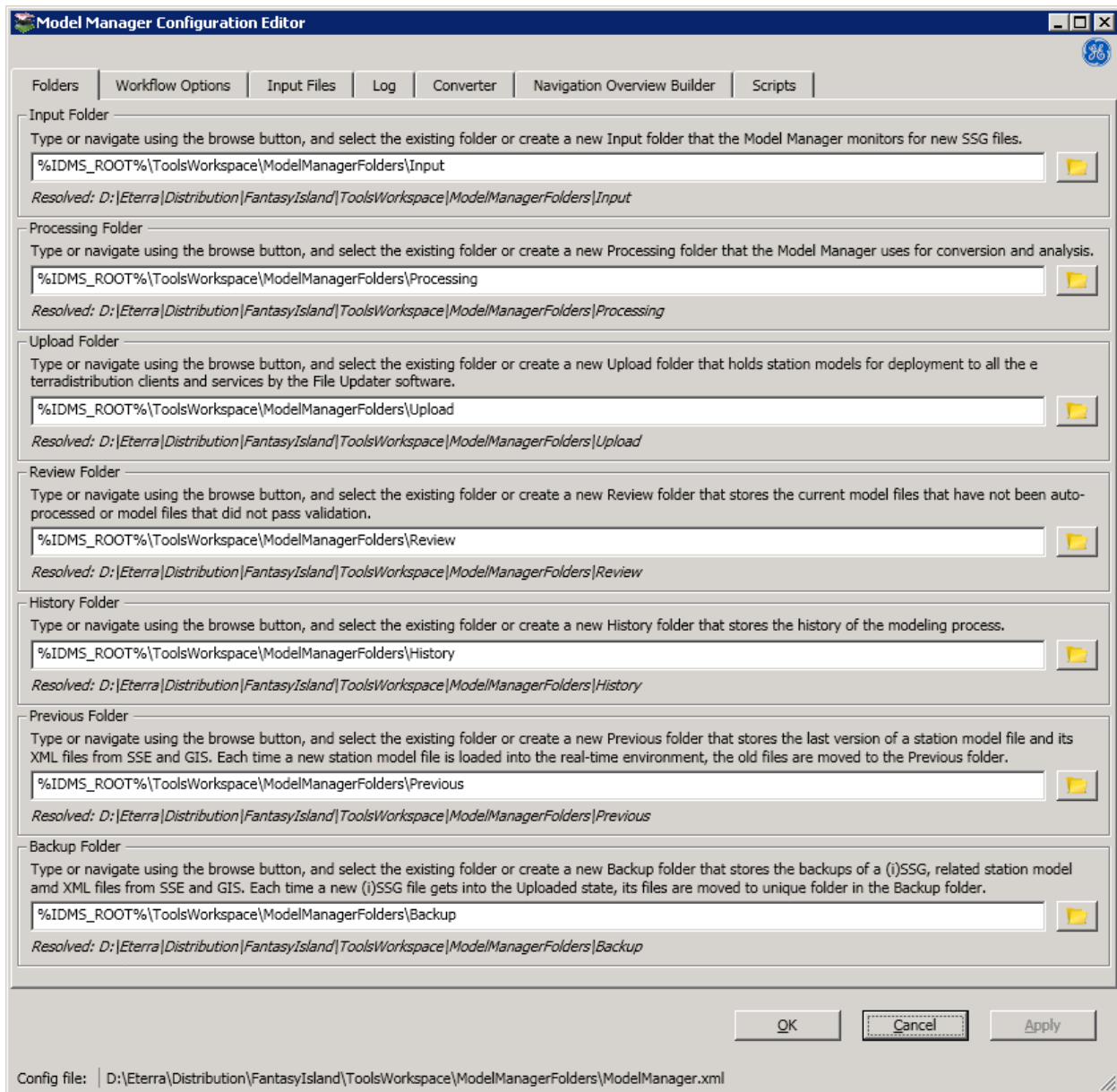


Figure 2. Model Manager Configuration Editor

The Model Manager Configuration Editor contains the following:

- **Folders tab:** Allows configuring the Model Manager's folders location.
- **Workflow Options tab:** Allows you to enable or disable blocking stations from the (I)SSG file when processing results in a timeout.
- **Input Files tab:** Allows you to add and remove input files and define the folders to move the files into during processing.
- **Log tab:** Sets the location of the log file and manages log message storage.
- **Converter tab:** Allows configuring the Converter executable and configuration file location and the number of workflows that the Converter can process simultaneously.

- **Navigation Overview Builder tab:** Allows configuring the location of the Navigation Overview Builder application, the number of stations that need to be built before the Navigation Overview Builder application runs, and the overview build idle delay.
- **Scripts tab:** Allows managing scripts used by the Model Manager to back up files, update a customer file with an incremental file, perform command-line operations, etc. A script object monitors system events and other events, to which a PowerShell script can be attached. The attached PowerShell script runs when the monitored event occurs.

For more information about the Configuration Editor options, refer to the *Model Manager User Guide*.

2.4. Model Manager Standalone Application Mode

The Model Manager standalone mode is used when the application is not required to run as a service and for testing purposes. The standalone Model Manager is installed with a default shortcut and configuration suitable to start most projects.

2.4.1. Configuring the Windows Shortcut

The Model Manager program should start in the same location as its executable. The Model Manager's shortcut passes a configuration file to the application.

You can access the shortcut from the Windows Start menu> **Eterra> Distribution> Tools> Modeling Tools> Model Manager Application**. This shortcut allows you to start the Model Manager standalone application. You can modify parameters via the Target and Start In fields of the shortcut's properties dialog box:

```
Target = D:\Eterra\Distribution\Tools\bin\ModelManagerApplication.exe /CONFIG
"%IDMS_ROOT%\ToolsWorkspace\ModelManagerFolders\ModelManager.xml"
```

```
Start in = D:\Eterra\Distribution\Tools\bin\
```

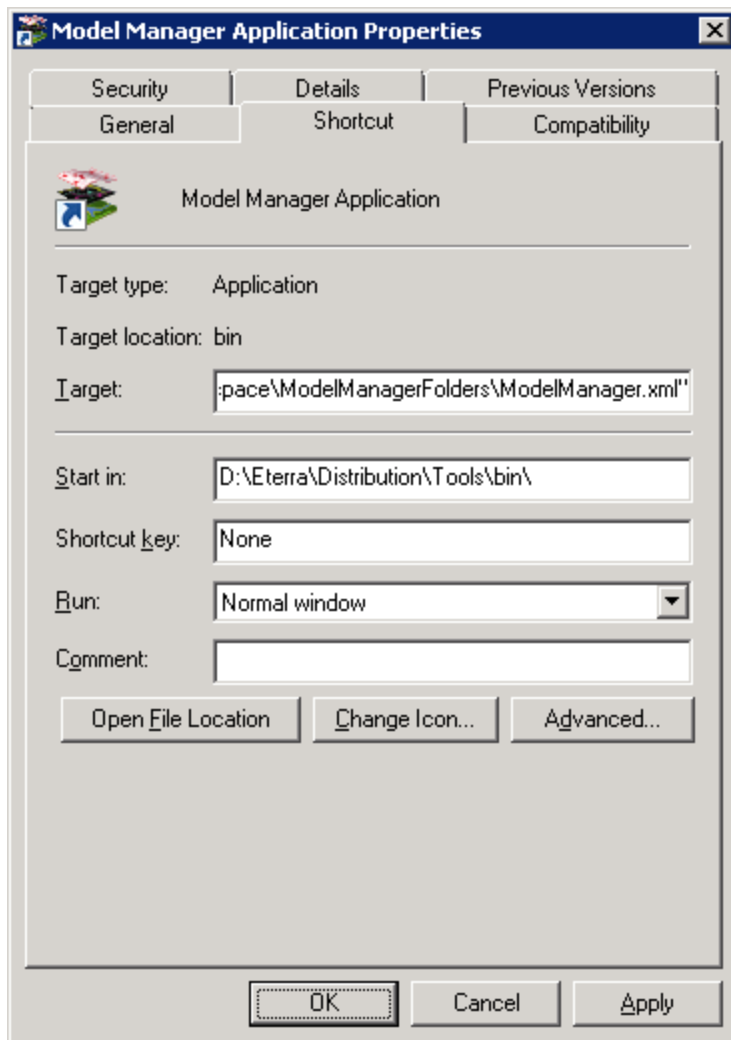


Figure 3. Model Manager Properties

2.4.2. Verifying a Standalone Installation

To verify a standalone installation:

1. Launch the Model Manager Application from the Windows start menu: **Eterra> Distribution> Tools> Modeling Tools> Model Manager Application.**
2. Place a sample SSG file in the Input folder as specified in the Input Files tab of the Model Manager Configuration Editor.
3. In the Model Manager application, navigate to the Log tab and verify that activity is triggered for this file.

The activity records indicate that basic configurations are correct.

Substation Group Files Summary			
History by Substation			
Upload History			
Log			
Date and Time	Type	User	Message
7/1/2015 10:53:14 AM	Info		Converting RAVEN
7/1/2015 10:53:14 AM	Info		Modeling workflow queued: RAVEN_001.SSG State: Converting
7/1/2015 10:53:13 AM	Info		File processed: D:\ETERRA\Distribution\FANTASYISLAND\ToolsWorkspace\ModelManagerFolders\INPUT\RAVEN_001.XML Moved file to processing: RAVEN.XML Copied to Review: RAVEN.XML File Owner: BUILTIN\Administrators
7/1/2015 10:53:13 AM	Info		File processed: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelManagerFolders\Input\RAVEN_001.SSG Moved file to processing: RAVEN.SSG Copied to Review: RAVEN.SSG File Owner: BUILTIN\Administrators
7/1/2015 10:53:13 AM	Info		Moving files into processing environment for RAVEN_001.SSG
7/1/2015 10:53:13 AM	Info		Modeling workflow queued: RAVEN_001.SSG State: MovingToProcessing
7/1/2015 10:53:12 AM	Info		Modeling workflow queued: RAVEN_001.SSG State: VerifyingInputData
7/1/2015 10:53:11 AM	Info		Modeling State Changed: Modeling
7/1/2015 10:53:11 AM	Info		Modeling workflow queued: RAVEN_001.SSG State: Queued
7/1/2015 10:53:10 AM	Info		Ssg file detected: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelManagerFolders\Input\RAVEN_001.ssg File Owner: BUILTIN\Administrators Last Date modified: 6/26/2014 8:27:09 AM
7/1/2015 10:51:07 AM	Info		0 logs retrieved from D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelManagerFolders\ModelManager.log
7/1/2015 10:51:07 AM	Info		New log file created because none was found.
7/1/2015 10:51:07 AM	Info	hab06	ModelManagerApplication (3.4.0.6530) launched
7/1/2015 10:51:06 AM	Info		Configuration loaded from: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelManagerFolders\ModelManager.xml

- For more comprehensive tests, place additional file types into the Input folder, including a station group file and station external and internal files. Once the files are copied, the Model Manager invokes the Converter and places the output .mod files into the Upload or Review folder.

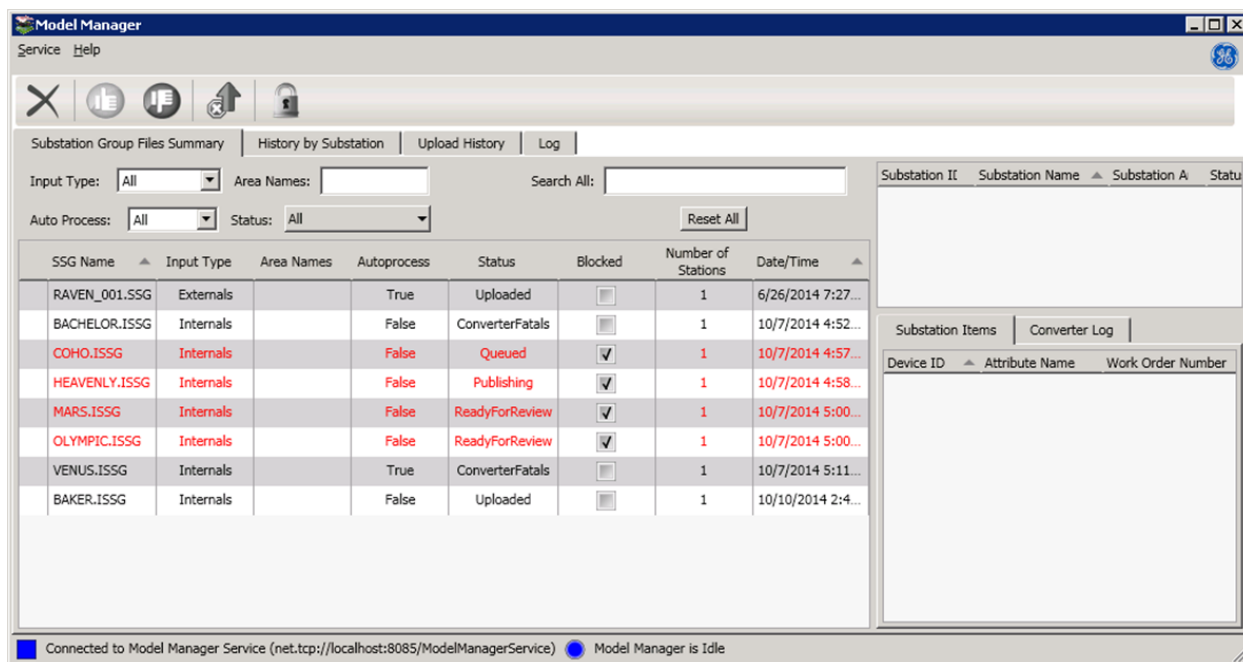


Figure 4. Standalone Model Manager Application Processing Input Files

2.5. Model Manager Service Mode

In Service mode, the Model Manager runs as a Windows service and constantly scans for new files. As files appear in the Input folder they are automatically processed and uploaded to the File Updater for transfer to the ADMS server. The analyst can review the files and log records and control the process using the Service Client.

The Windows Service application, `ModelManagerService.Provider.exe`, runs 24x7 under the `NetworkService` account (or another account, as configured). This application includes a self-hosted web server, which also requires configuration. Multiple users on the same host can launch the client `ModelManagerService.Client.exe` and connect to the service simultaneously. The user interface of the client instance is kept in sync with the service. Currently only the localhost connection is supported.

The following sections describe configuring the Windows Service provider and client.

2.5.1. Windows Shortcuts

You can access the Model Manager Service shortcuts from the Windows Start menu > **Eterra** > **Distribution** > **Tools** > **Modeling Tools**. The shortcuts allow you to start and stop the Model Manager Service, start the Model Manager Service Client, and run the Model Manager Standalone application.

The following shortcuts are available:

-

Start Model Manager Service

- Stop Model Manager Service
- Model Manager Service Client

2.5.2. Installing the Service Provider

The tools kit installation process registers the `ModelManagerService.Provider.exe` file with the Windows Service host and labels it "ADMS Model Management Service". The service can be controlled and additionally configured using the Windows Services application.

To open the Windows Services application:

- Navigate to the Control Panel> System and Security> Administrative Tools> Services.

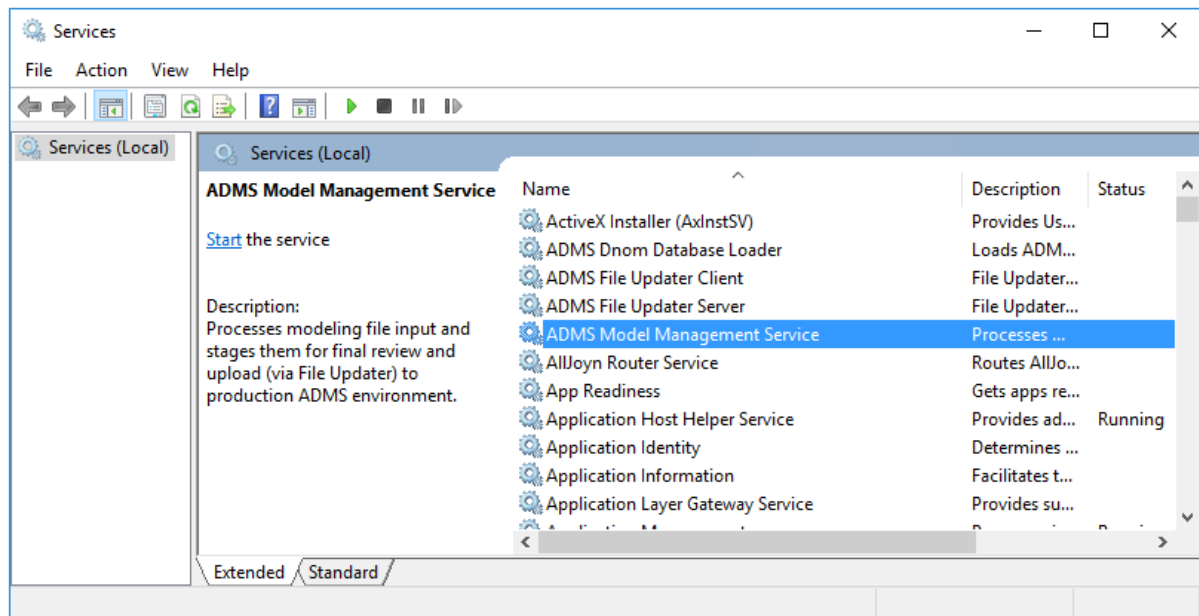
Or

- Run `services.msc` from the Windows Start menu or command line.

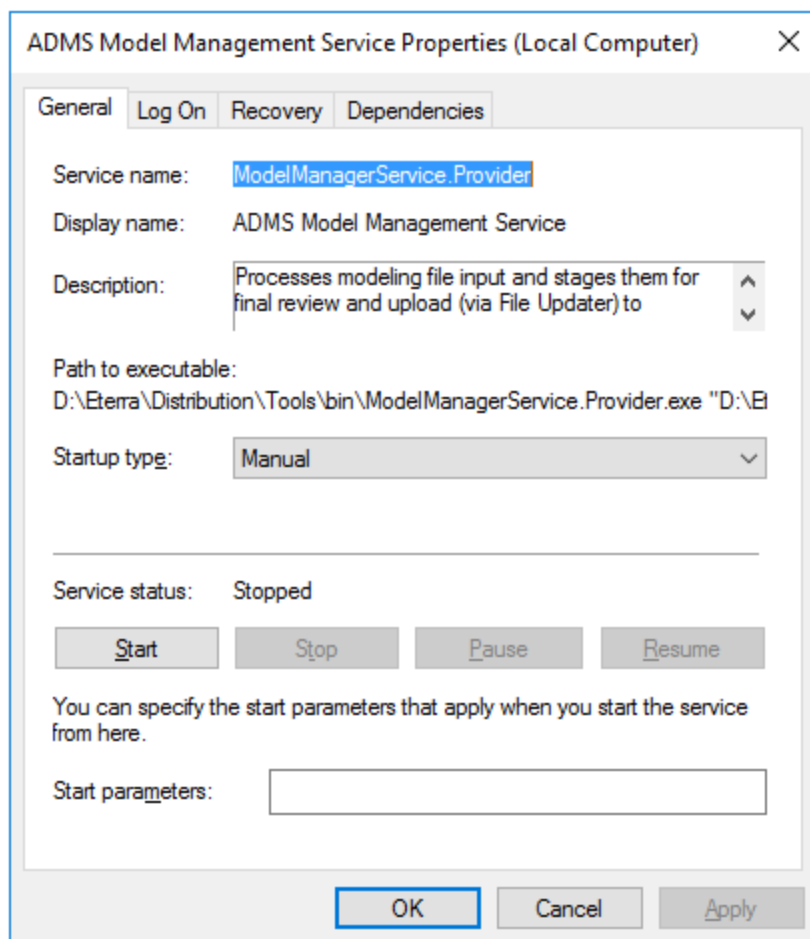
2.5.2.1. Verifying Installation

To verify that the Service Provider is installed properly:

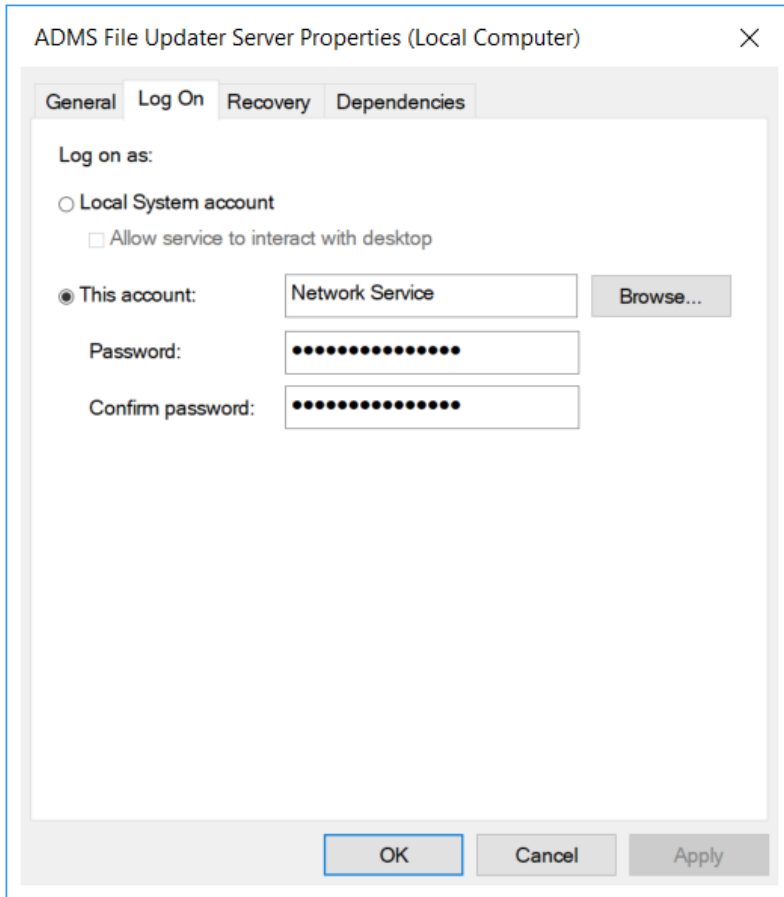
1. Run `services.msc` from the Windows Start menu or command line. The Service Management Console opens.
2. Locate the ADMS Model Manager Service.



3. Right-click the Service and select Properties. The Properties dialog box appears.



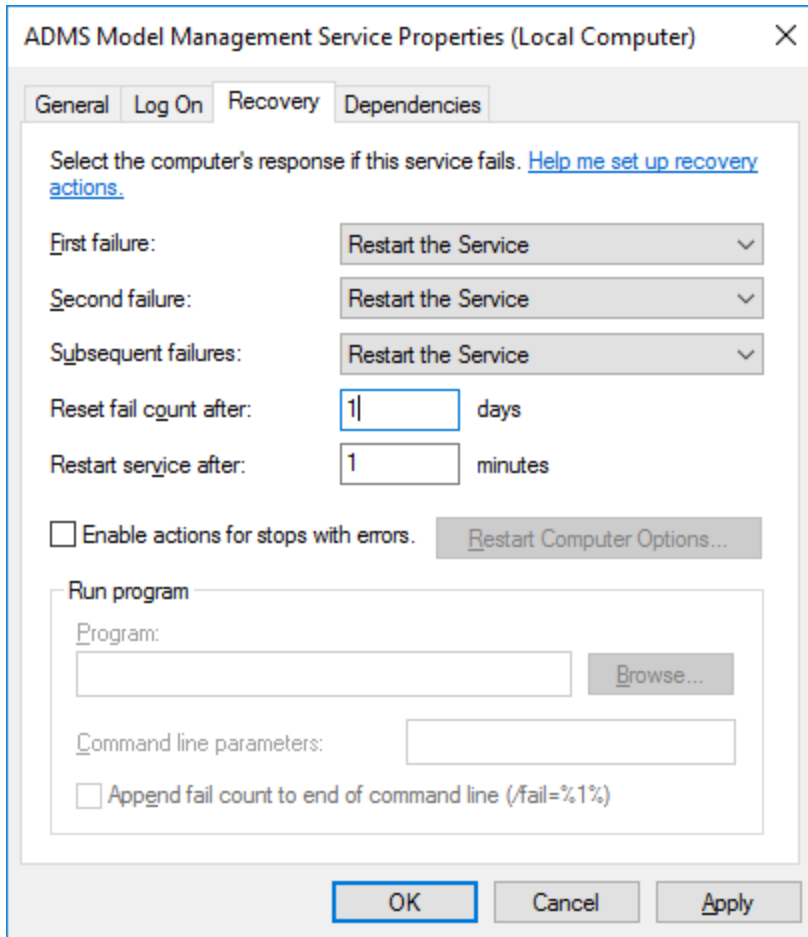
4. On the General tab, in the Path to Executable field, verify the command line arguments.
 - The first argument is the path to the executable:
`D:\Eterra\Distribution\Tools\bin\ModelManagerService.Provider.exe`
 - The second argument is the path to the configuration file:
`D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\ModelManagerFolders\ModelManager.xml`
 - The third argument is the service endpoint URL:
`net.tcp://localhost:8085/ModelManagerService`
5. If needed, on the General tab, set Startup Type to Automatic via the Startup Type drop-down list.
6. On the Log On tab, verify that "This account" is selected with "Network Service" account.



Note

By default, the Model Manager service always logs on as "Network Service".

7. On the Recovery tab, configure the recovery actions to match the example below:



8. Stop and start the service.

2.5.2.2. Reviewing Model Manager's Event Log

If the service was not started successfully:

1. Start the Windows Event Viewer:
 - **Control Panel> System and Security> Administrative Tools> Event Viewer**
- Or
 - From the Windows Start menu or command line, run `eventvwr.msc`.
2. In the Event Viewer's left panel, expand Applications and Services Logs and select Model Manager Event Log.
3. Select the latest log entry to view any error conditions that may be preventing the Model Manager Service Provider from running.

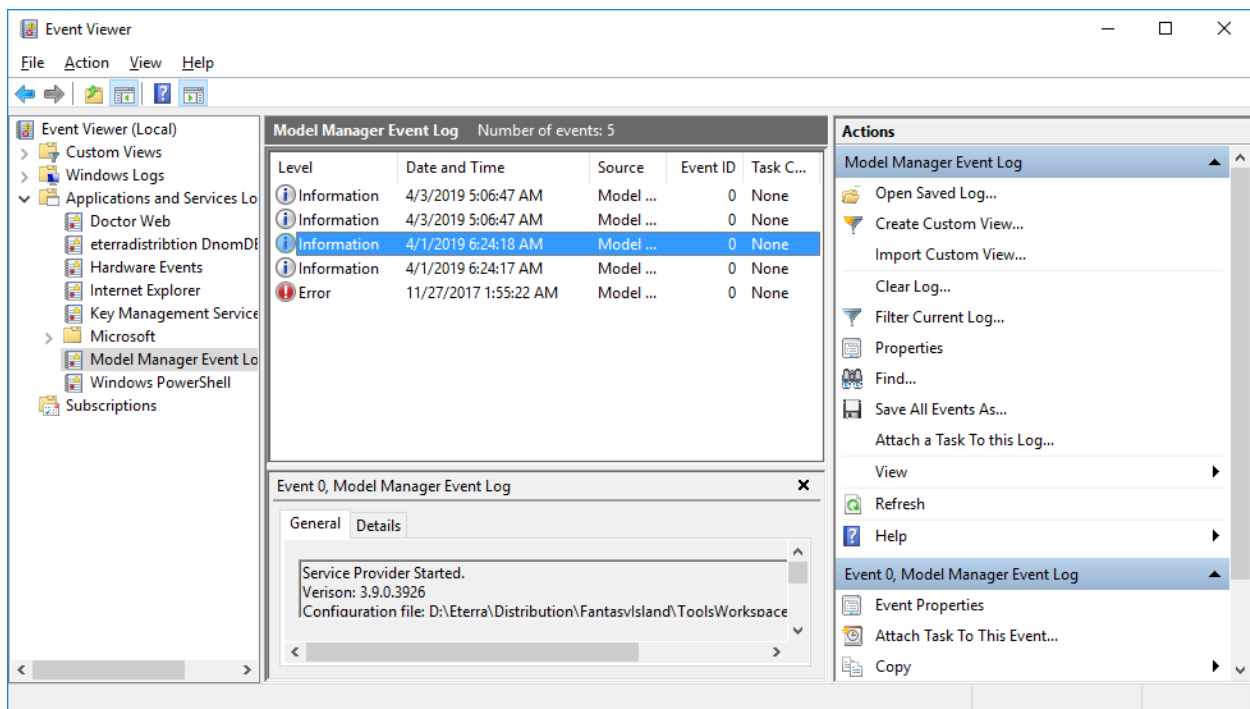


Figure 5. Windows Event Viewer Displaying Model Manager Event Log

2.5.3. Delegating Permissions for Non-Admin User to Manage Windows Service

An administrator can optionally provide any user permission to manage a Windows Service, including stopping and starting the Service Provider.

For more information, refer to: <http://support.microsoft.com/kb/914392>.

2.5.4. Starting the Client and Verifying the Setup

Alter the ModelManagerApplication.exe standalone application configuration is successfully verified, you can verify the installation and setup of the Service components.

To start the Model Manager Service Client, from the Windows Start menu, navigate to **Eterra> Distribution> Tools> Modeling Tools> Model Manager Service Client**.

When the Service Client starts, the Status Indicator at the bottom of the display indicates if it is connected to the Service Provider.

 Connected to Model Manager Service (net.tcp://localhost:8085/ModelManagerService)

View the Log tab for any potential issues.

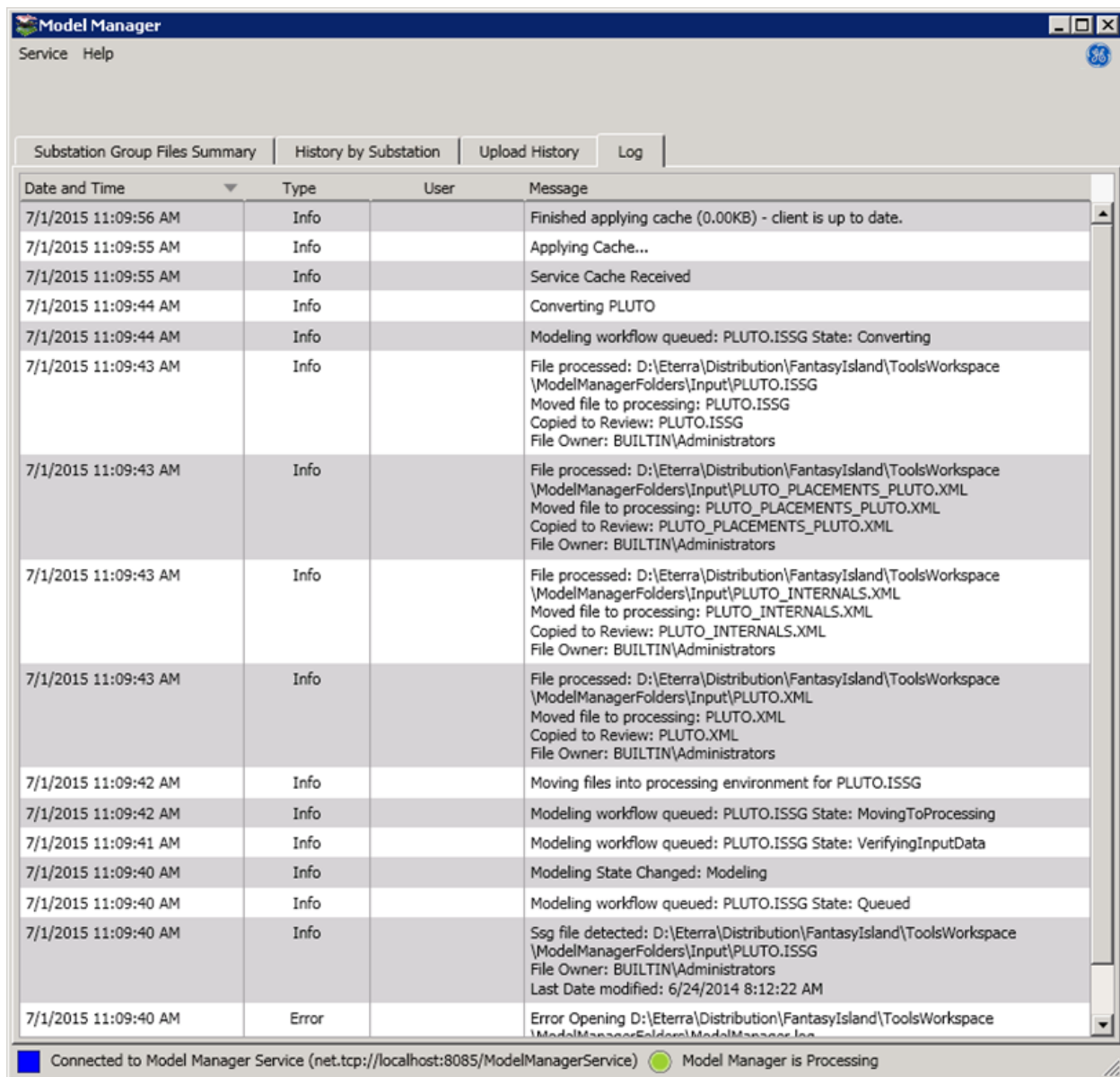


Figure 6. Model Manager Service Client Connected to Service Provider

2.5.5. Verifying Startup and Other User Access

To verify the client can connect correctly using a different user account:

1. Reboot the machine.
2. Log in as another user.
3. Verify the Service is running.
4. Start the client and verify the setup as described above.

2.5.6. Verifying Model Manager Processing

To comprehensively test the Model Manager configuration, run sample data using the following steps:

1. Create one or more station placement files with the Substation Editor (for steps, refer to the *Substation Editor User Guide*).
2. Copy the station files including `<station>.xml`, `<station>_subeditor.xml`, `<station>.ssg`, and `<station>.issg` from the sample `ToolsWorkspace\Converter\Input` directory into the `Input` directory of the Model Manager (`\ToolsWorkspace\ModelManagerFolders\Input`).
3. In the Model Manager (standalone application or service client), view the progress as described in the *Model Manager User Guide*.

Verify that the Log tab contains no error messages.

The next phase of configuration and validation involves validating that the File Updater moves the output of the Model Manager, including `.mod` files placed in its `Upload` folder, onto the ADMS server.

3. Model Manager Web Service

The Model Manager web service synchronizes electric distribution networks among the Model Manager and the Smallworld Electric Office GIS Adapter (GISA). It helps exchange SOAP messages and keep the substation models in sync.

To provide security for exchanged SOAP messages, the HTTP Transport Layer Security (TLS) protocol with mutual certificate authentication is used. TLS is a security protocol that is based on the Secure Sockets Layer (SSL) protocol. TLS uses digital certificates to authenticate users as well as authenticate the network.

3.1. Accepting SOAP Messages from GISA

To accept SOAP messages from GISA over a secure channel, a self-hosted WCF Model Manager web service is configured with an SSL (X.509) certificate. This allows a GISA client to verify the identity of the Model Manager (MMA) server. The GISA client is also configured with an X.509 certificate for the MMA web service to verify the identity (client authentication) of the GISA client. The MMA server's certificate must be trusted by the GISA client, and the GISA client's certificate must be trusted by the MMA server.

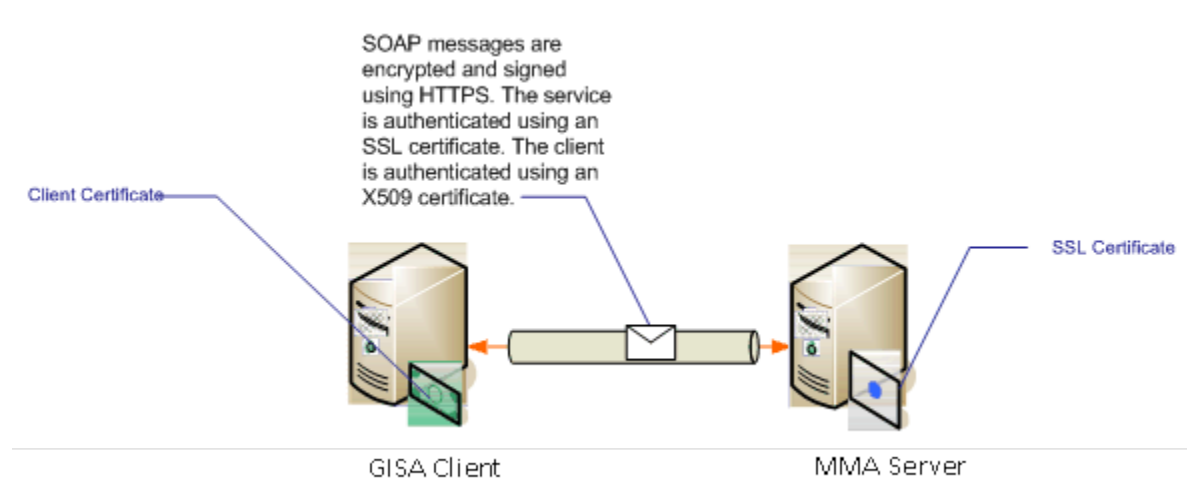


Figure 7. TLS with Certificate Authentication among a GISA Client and an MMA Server

To enable HTTPS on a web server and to enable client authentication, you need to have a certificate. In production environments, for security reasons, certificates must be signed by a Certificate Authority (CA). For testing purposes, you may generate self-signed certificates.

3.1.1. Generating and Configuring a Self-Signed Certificate

Warning

Self-signed certificates created using the instructions in this section should only be used in a

product testing environment.

3.1.1.1. Creating a Self-Signed Certificate

To create a Model Manager Server self-signed certificate using makecert.exe:

1. Open a Command Prompt window as an administrator.
2. Change to the `C:\Program Files (x86)\Windows Kits\8.1\bin\x64` directory.
3. Verify that you can run the makecert command.

Note

If the makecert application is not available on your [Windows Server computer](#), you can use another Windows computer where the makecert tool is installed. In this case, export your certificates with the private key option, and then import them on the server.

4. Run the following command:

```
makecert -pe -r -n "CN=MMASoapInterface" -len 2048 -sky exchange -eku 1.3.6.1.5.5.7.3.1 -ss My -sr LocalMachine -m 240 MMASoapInterface.cer
```

Where:

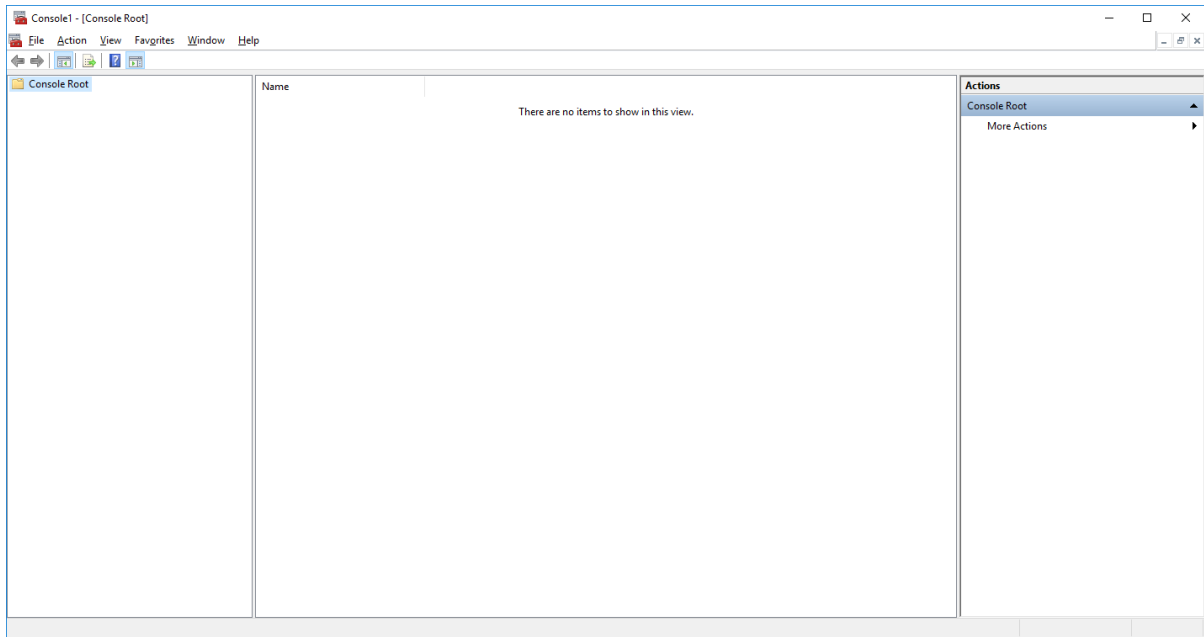
- **pe:** Marks a generated private key as exportable.
- **r:** Creates a self-signed certificate.
- **n:** Certificate subject name. Use the "MMASoapInterface" subject name.
- **len:** Generated key length (in bits). Default values are 2048 for RSA and 512 for DSS.
- **sky:** Subject key type (signature, exchange, or integer).
- **eku:** Comma-separated enhanced key usage OIDs.
- **ss:** Subject's certificate store name that stores the output certificate.
- **sr:** Subject's certificate store location (CurrentUser or LocalMachine). Default value is CurrentUser.
- **m:** The certificate validity period, in months.

3.1.1.2. Configuring a Self-Signed Certificate

To configure a self-signed certificate for use by the Model Manager web service:

1. Open the Microsoft Management Console application. To access this application, enter `mmc` in the Windows Start menu.

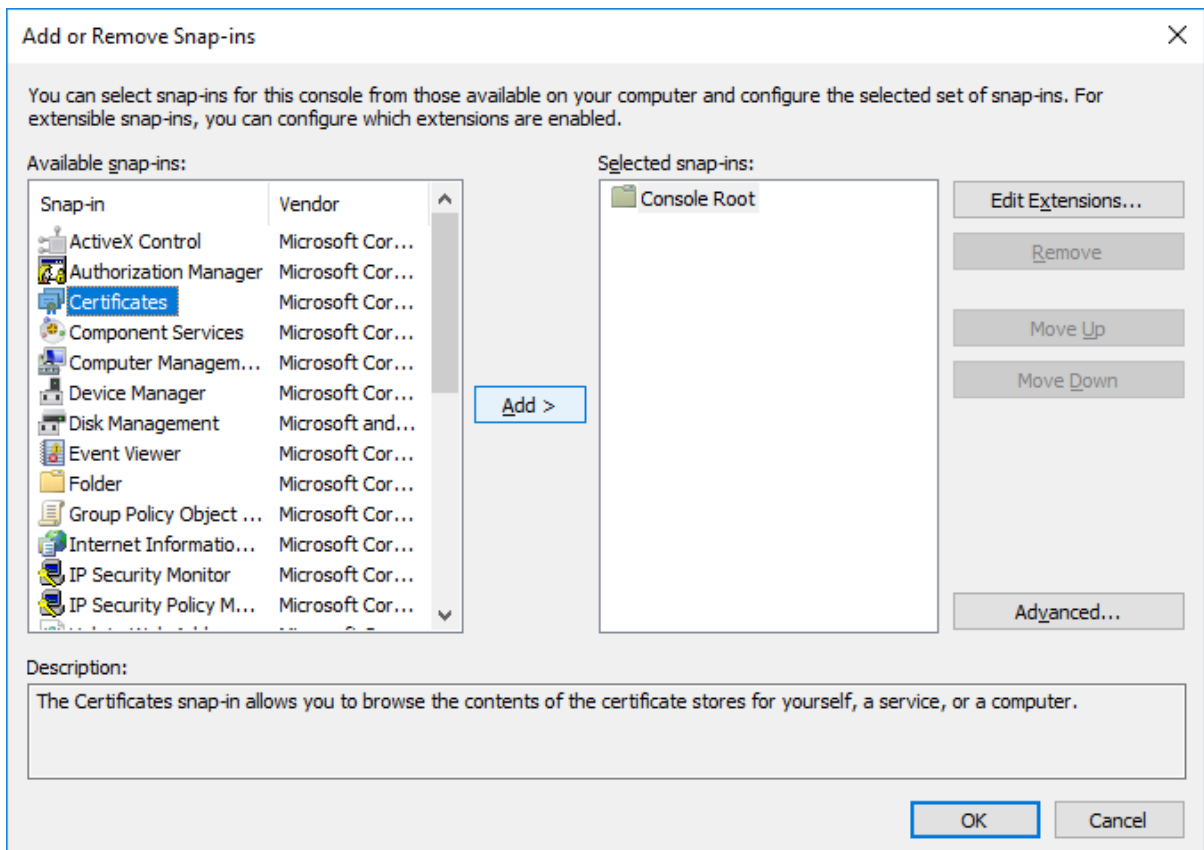
The Console Root location shows all certificates on the local computer.



2. From the File menu, click Add/Remove Snap-in.

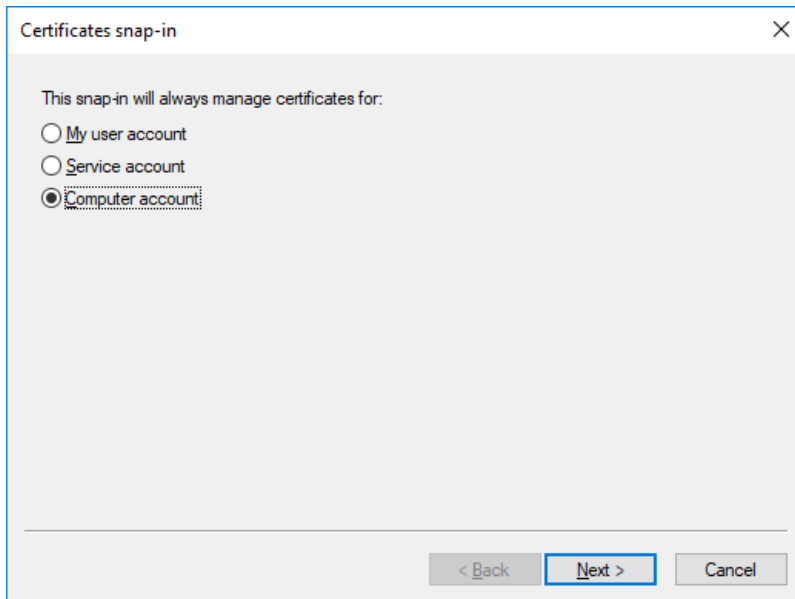
The Add or Remove Snap-ins window appears.

3. In the "Available snap-ins" group, select Certificates, and then click Add.

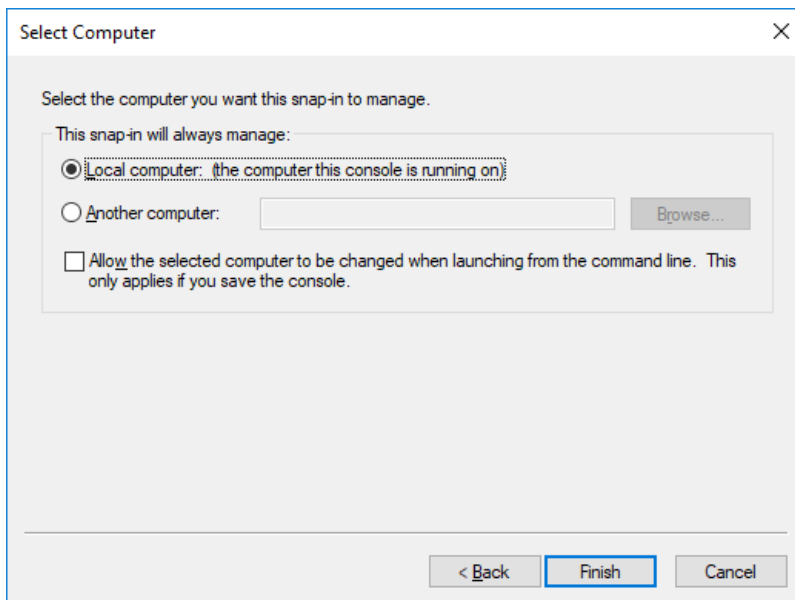


The Certificates snap-in window appears.

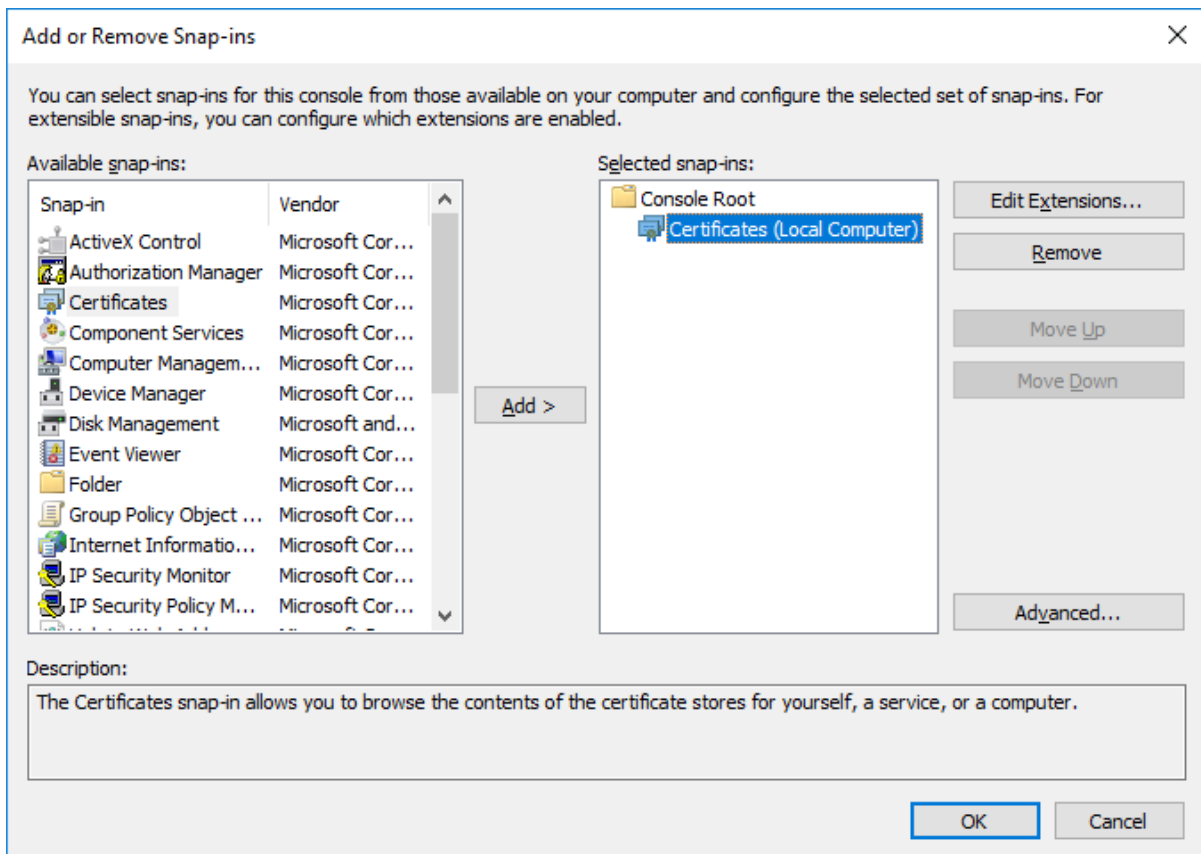
4. Select "Computer account" and click Next.



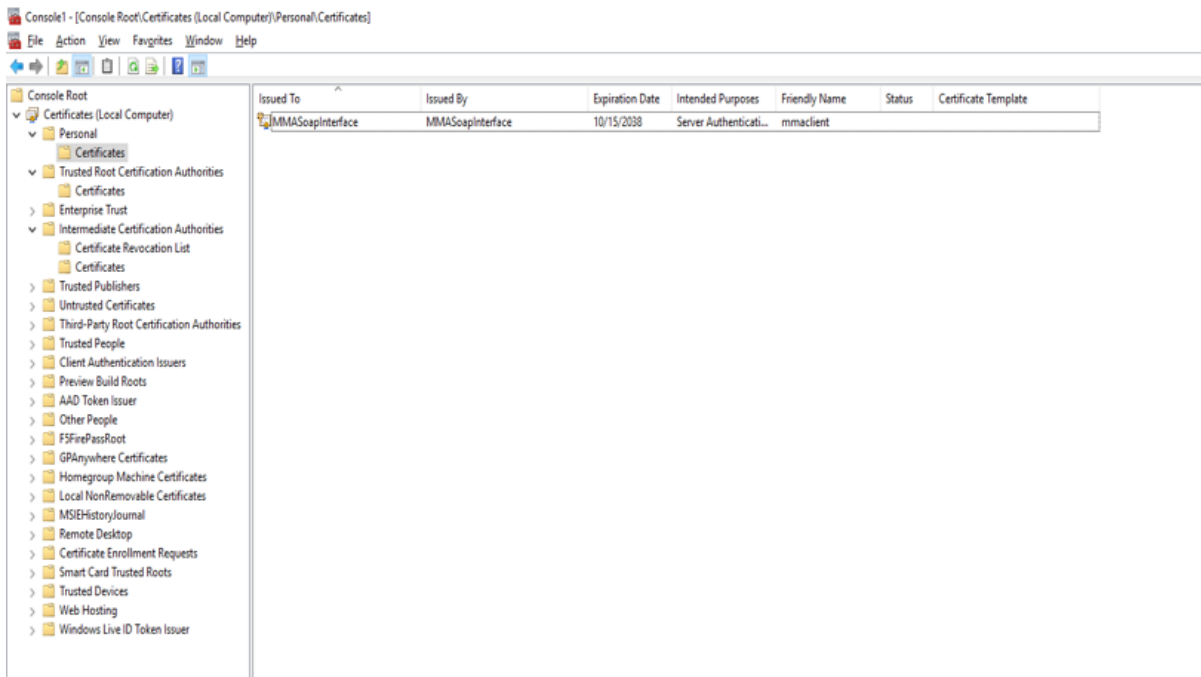
5. Select "Local Computer" and click Finish.



6. Verify that the Certificates (Local Computer) element appears under Console Root in the "Selected snap-ins" group. Click OK.

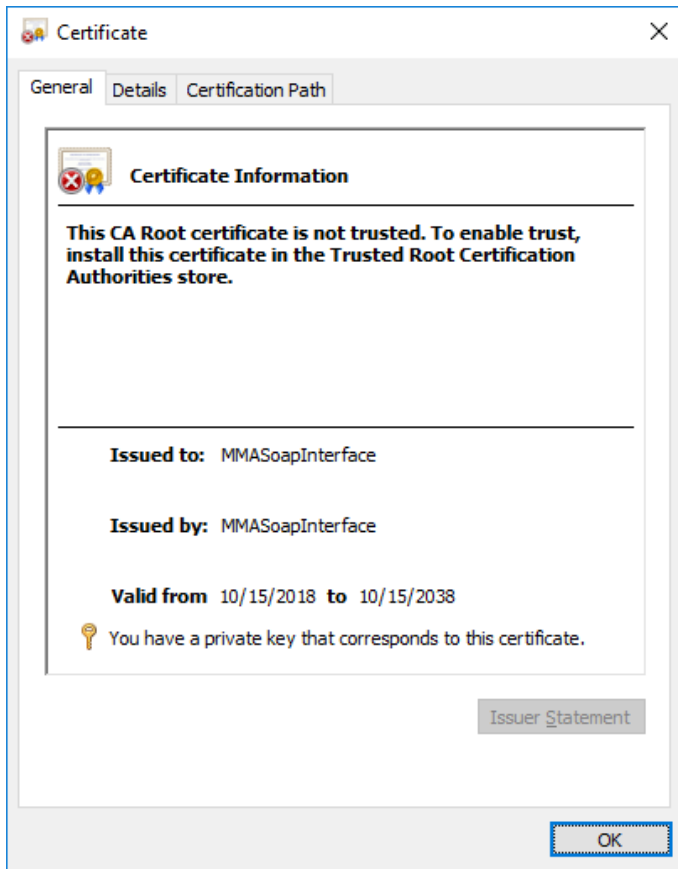


7. Verify that the new certificate appears in the Personal > Certificates folder and that Server Authentication appears in the Intended Purposes column.

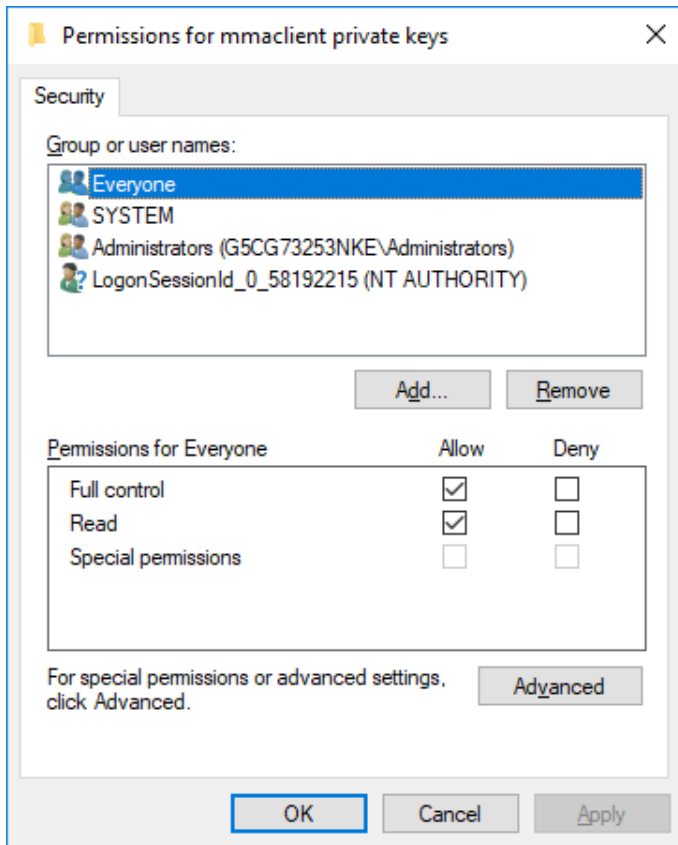


8. Double-click the certificate to view the certificate details.

The General tab shows that the certificate must be copied to the Trusted Root Certification Authorities > Certificates folder to become trusted.



9. Right-click the certificate, and from the All Tasks menu, click Manage Private Keys.
10. Add permissions for the Windows users on the machine to use private key certificates.



3.1.2. Adding SSL Binding

HTTP Transport Layer Security requires an SSL (Secure Sockets Layer) certificate to bind with a port where the web service is running.

To bind an SSL certificate to a port that is used by the MMA web service:

1. Open a Command Prompt window as an administrator.
2. Run the following command:

```
netsh http add sslcert iport=0.0.0.0:<port number> certstorename=MY certhash=<certificate Thumbprint> appid=\{C60EF341-984E-4262-AE0F-8DEBB6D98255}
```

For example:

```
netsh http add sslcert iport=0.0.0.0:8090 certstorename=MY  
certhash=9eea9a3d112a4df0ec120efba1f9e968255f4651 appid=\{C60EF341-984E-4262-AE0F-8DEBB6D98255}
```

Where:

- **Ipport:** IP address and port for the binding. If HTTP is being used, the port should be 8090. If HTTPS, the port should be 8091. The MMA configuration default is 8090.
-

Certstorename: Store name for the certificate. Required for Hostname-based configurations. Defaults to MY for IP-based configurations. The certificate must be stored in the local machine context.

- **Certhash:** The SHA hash of the certificate. This hash is 20 bytes and is specified as a hex string.
- **Appid:** GUID to identify the owning application. The GUID of the Model Manager application is C60EF341-984E-4262-AE0F-8DEBB6D98255.

To obtain a certhash, do the following:

1. Open the Microsoft Management Console application. To access this application, enter `mmc` in the Windows Start menu.
2. In the Microsoft Management Console, select the "MMASoapInterface" certificate from Certificates (Local Computer) > Personal > Certificates.
3. Open a certificate and switch to the Details tab.
4. Select the Thumbprint field and copy the hexadecimal string.
5. Open a text editor, such as Notepad++.
6. Set encoding to ANSI.
7. Paste the copied hexadecimal string.
8. Remove the first character (" ") and empty spaces.

This is the certhash.

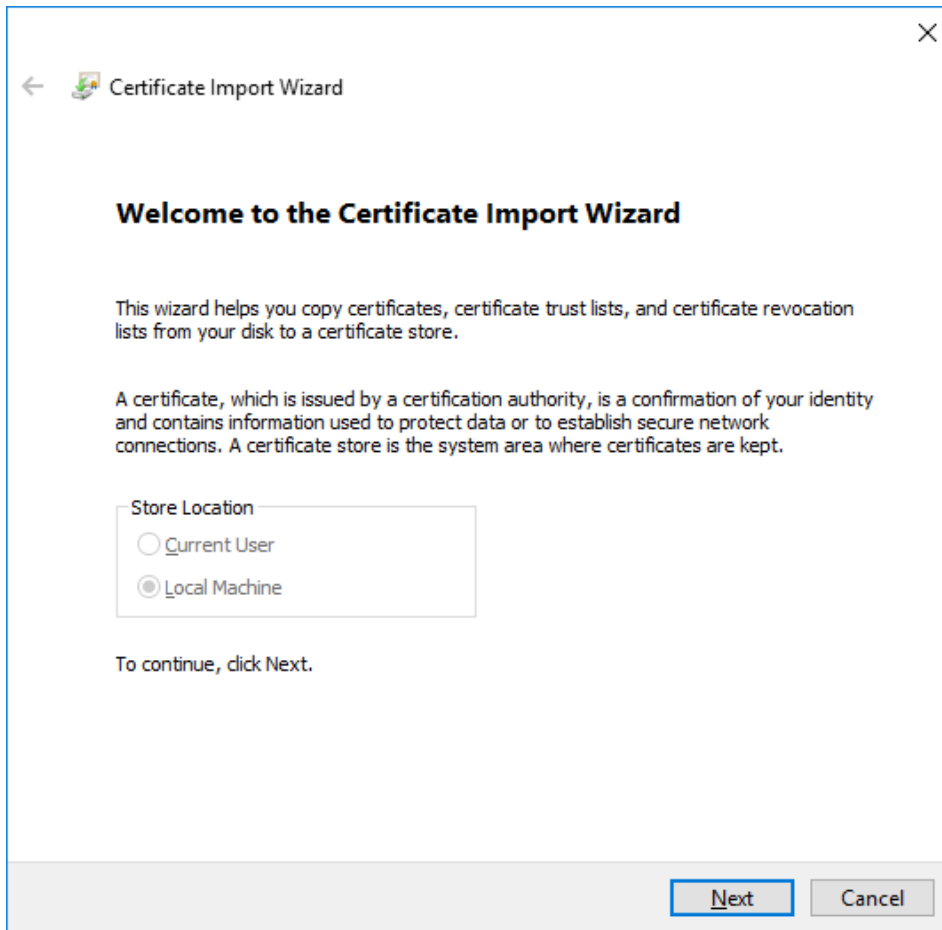
3.1.3. Importing a GISA Certificate into a Trusted Root CA

In client certificate authentication, clients are authenticated without a user's login and password. Instead, the server asks the client to show its certificate and verifies its identity. During the verification process, the server checks whether the client's certificate chains to a trusted root certificate authority.

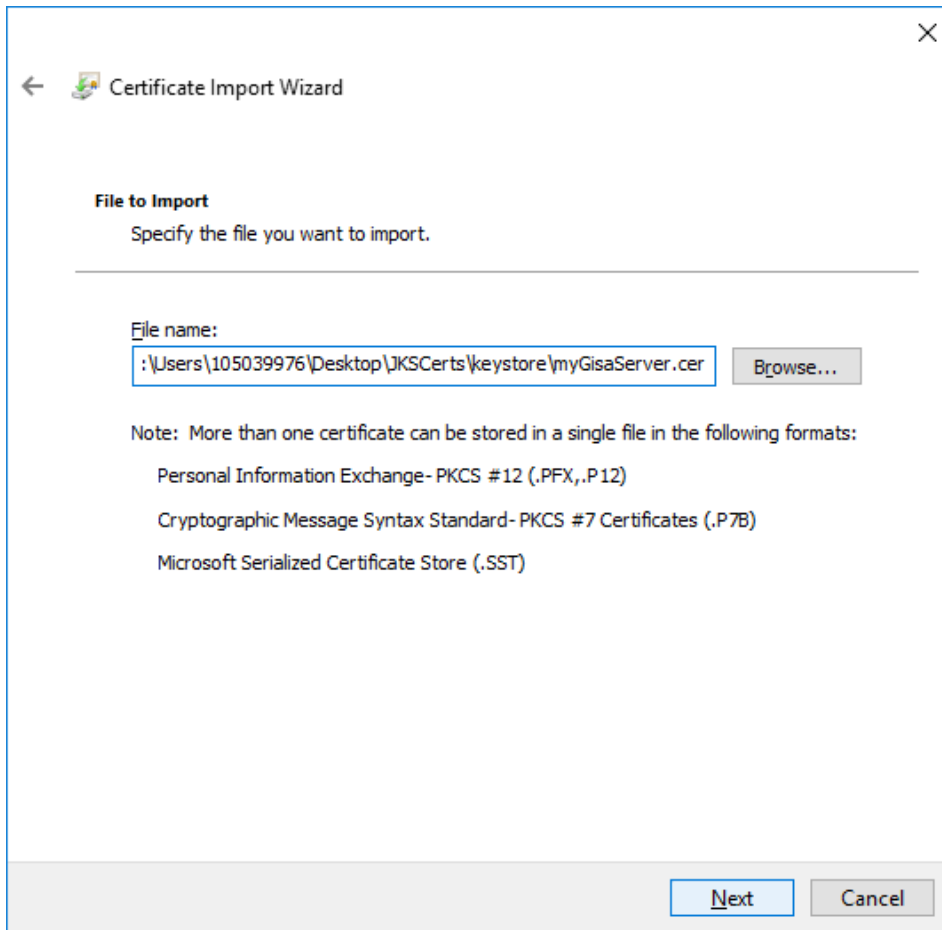
In the GISA-MMA interface, the MMA server checks the identity of the GISA client. For client authentication to be successful, the GISA client certificate needs to be imported into the Trusted Root CA folder.

To import a certificate into the Trusted Root CA folder:

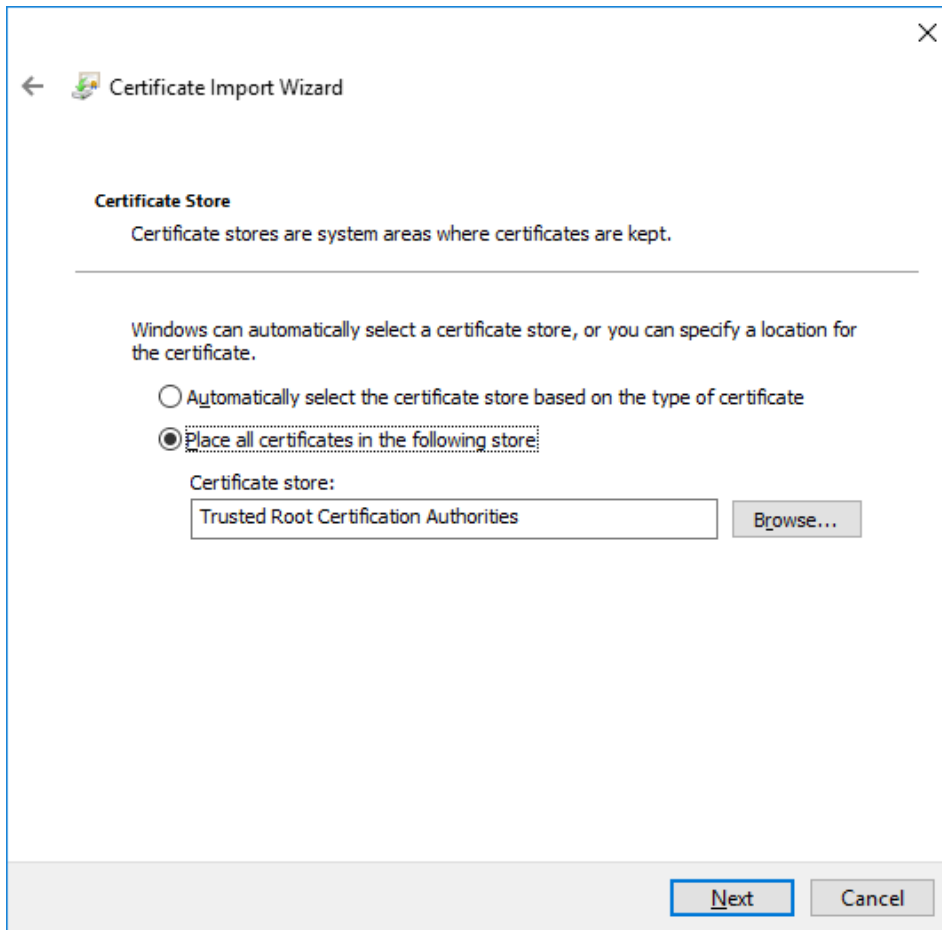
1. In Microsoft Management Console, open the Certificates folder under Console Root > Certificates (Local Computer) > Trusted Root Certification Authorities.
2. From the Actions menu, point to All Tasks, and click Import.
3. At the Certificate Import Wizard window, Click Next.



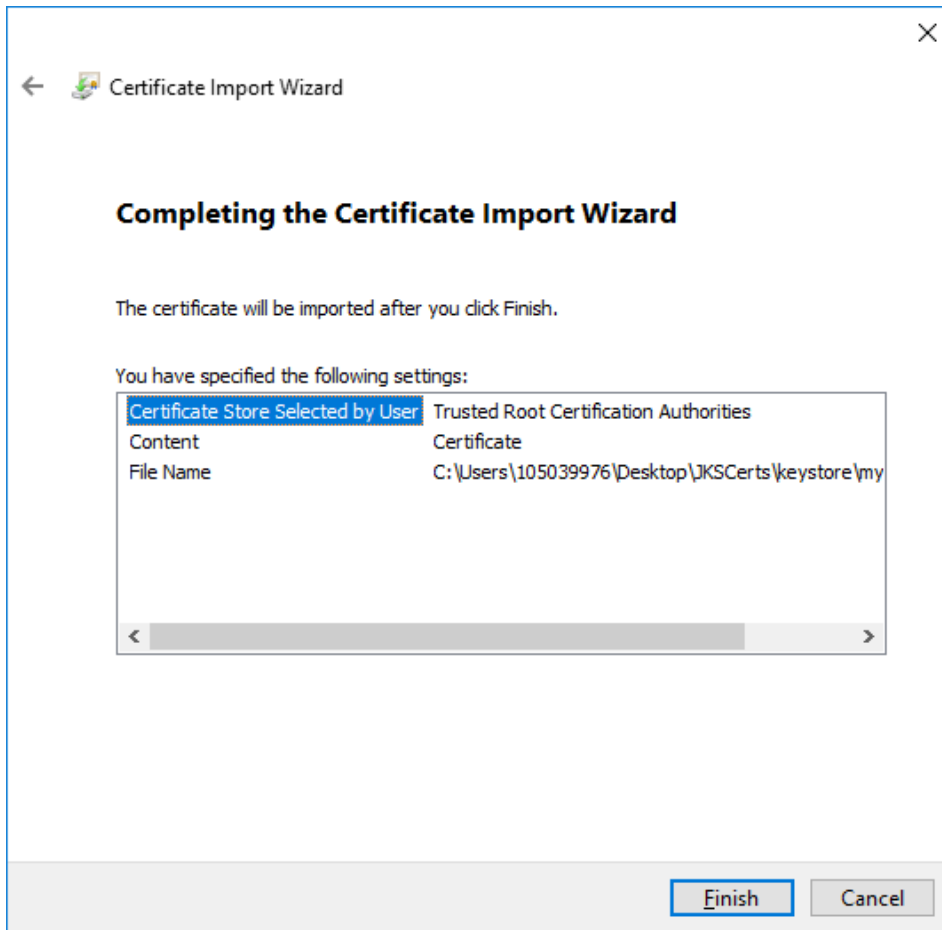
4. At the File to Import step, click Browse, locate the GISA certificate, and click Next.



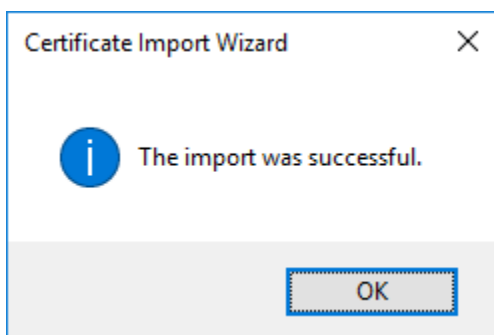
5. At the Certificate Store step, the "Place all certificates in the following store" option is selected by default. Click Next



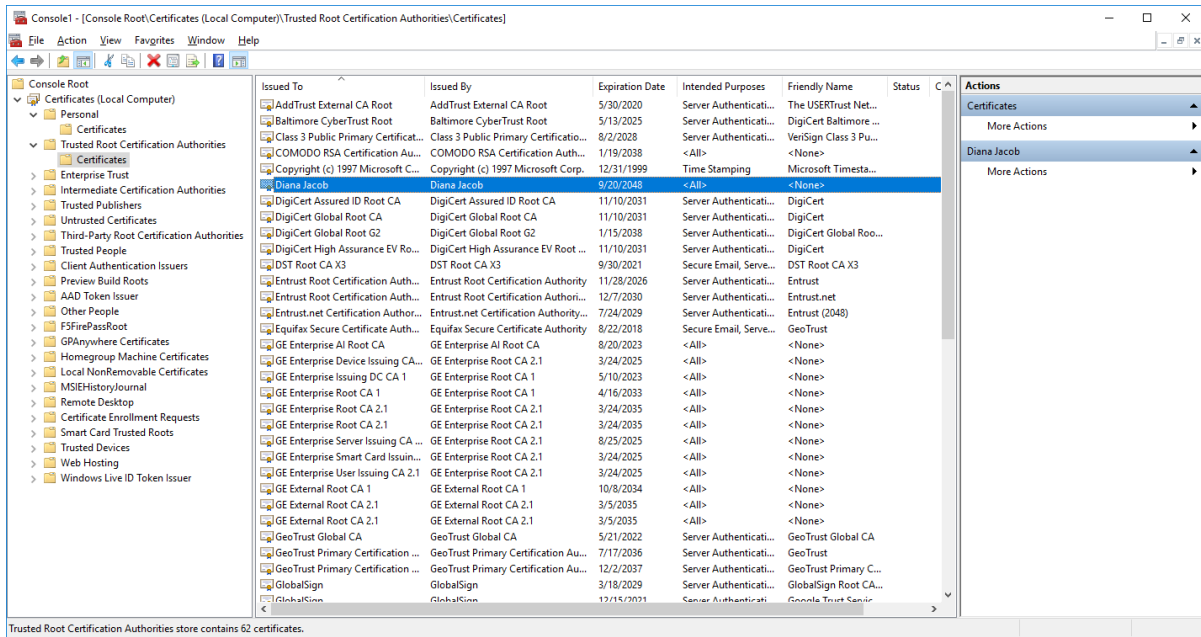
6. At the Completing the Certificate Import Wizard step, click Finish.



7. A message box appears indicating "The import was successful". Click OK.



8. Verify that the certificate was successfully imported to the Trusted Root Certification Authorities folder.



3.1.4. Adding a Namespace Reservation for the Port

A namespace reservation assigns the rights for a portion of the HTTP URL namespace to a group of users. A reservation gives those users the right to create services that listen on that portion of the namespace. Reservations are URL prefixes, meaning that the reservation covers all sub-paths of the reservation path.

To add a namespace reservation for the MMA self-hosted WCF web service:

1. Open a Command Prompt window as an administrator.
2. Run following command:

```
netsh http add urlacl url=https://+:<port number>/ user=<User Name>
```

For example:

```
netsh http add urlacl url=https://+:8090/ user=EVERYONE
```

3.2. Sending SOAP Messages to GISA

To send SOAP messages from GISA over a secure channel, the GISA (GSS) web service is configured with a Java Key Store along with an SSL (X.509) certificate that allows an MMA client to verify the identity of the GISA server. The MMA client is also configured with an X.509 certificate that allows the GISA web service to verify the identity (client authentication) of the MMA client. The GISA server's certificate must be trusted by the MMA client, and the MMA client's certificate must be trusted by the GISA server.

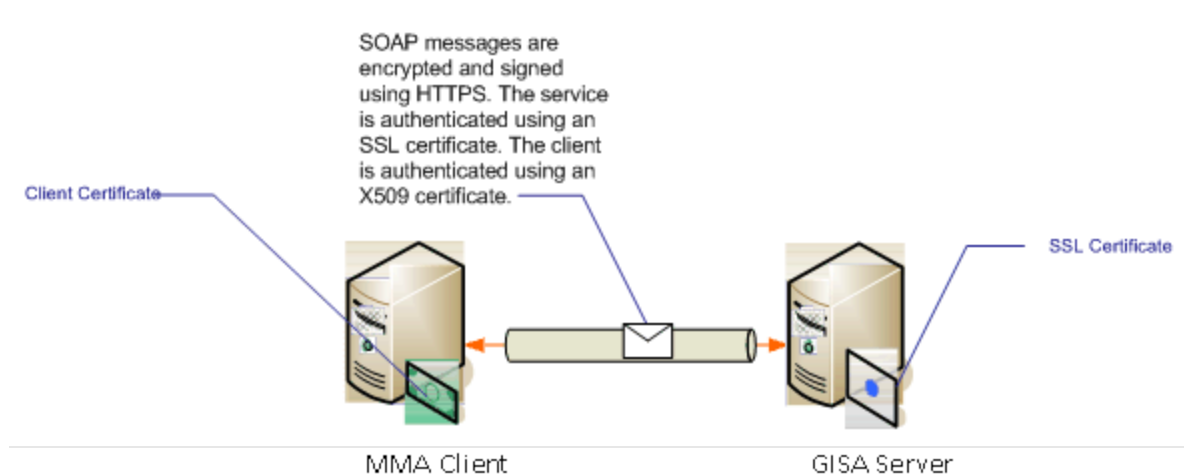


Figure 8. TLS with Certificate Authentication among the MMA Client and the GISA Server

Note

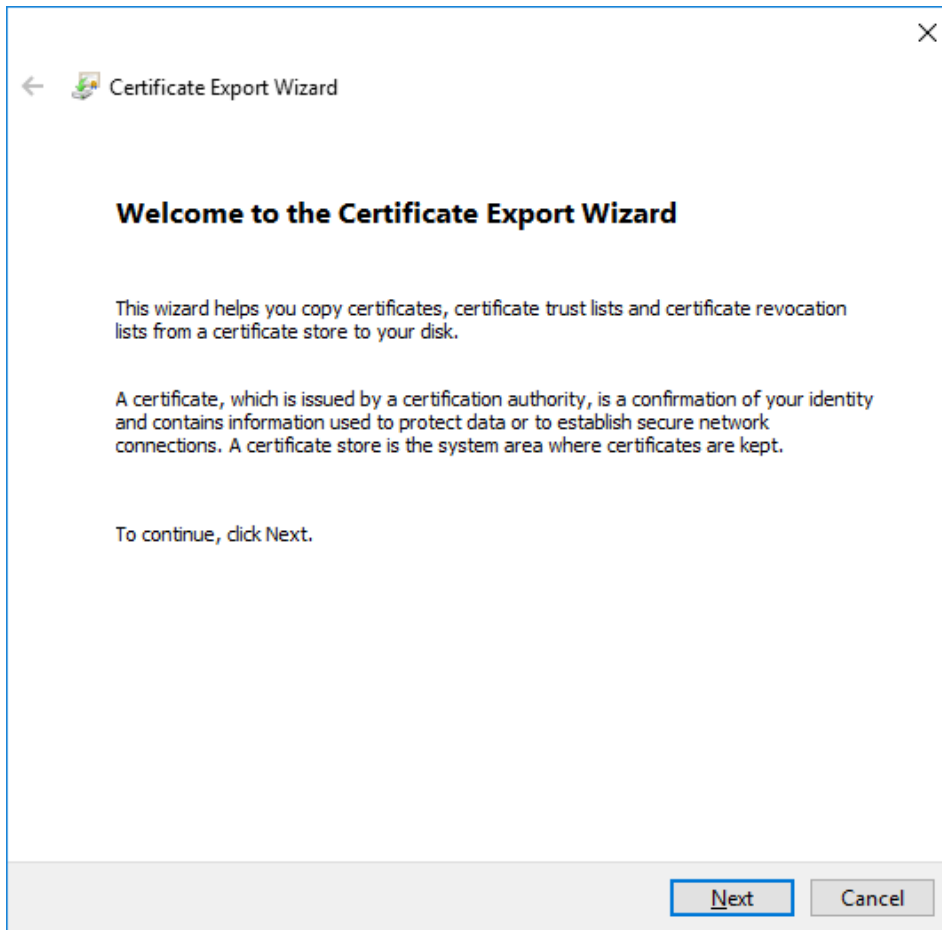
Currently, MMA uses only one SSL certificate for binding with a service port as well as for client authentication. GISA also uses only one SSL certificate for binding with a service port as well as for client authentication.

3.2.1. Exporting an MMA Server Certificate for GISA Client Usage (Export without Private Keys)

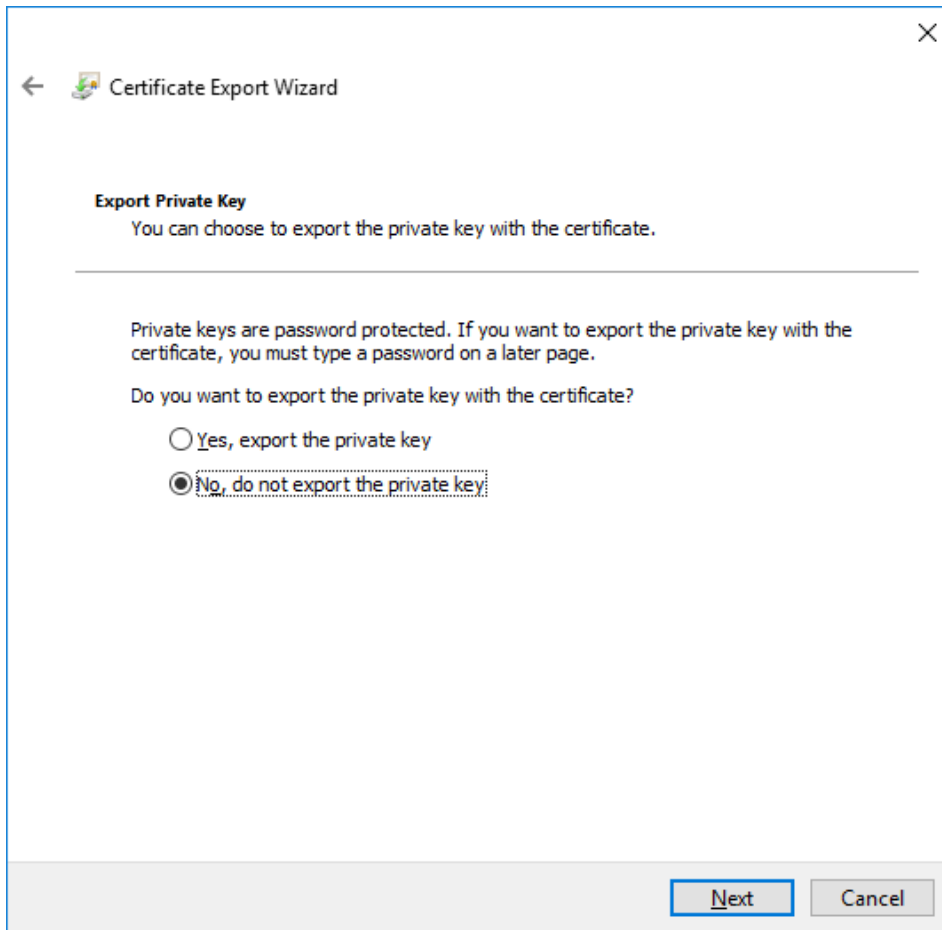
The MMA client's SSL certificate needs to be exported from the MMA client and then imported to the GISA server's Trusted Root Store.

To export an SSL certificate from the MMA client:

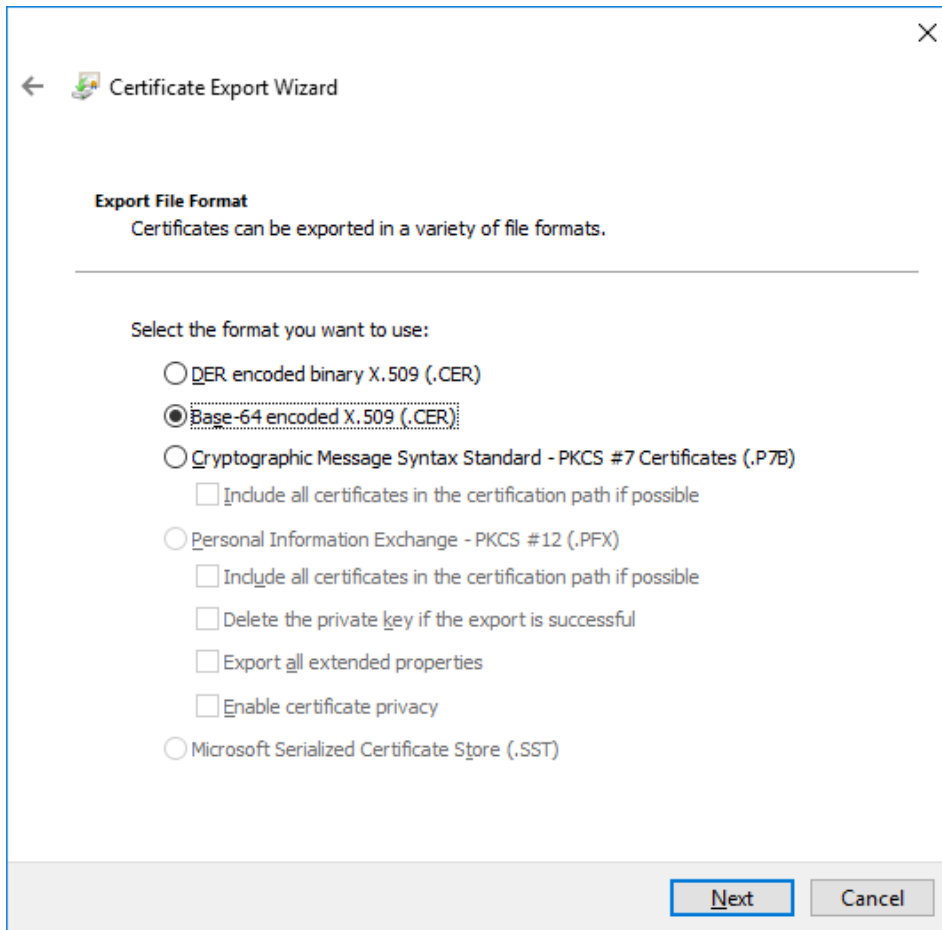
1. In Microsoft Management Console, select the "MMASoapInterface" certificate from the Personal > Certificates folder.
2. Right-click the "MMASoapInterface" certificate, and from the All Tasks menu, click Export.
3. At the Certificate Export Wizard window, click Next.



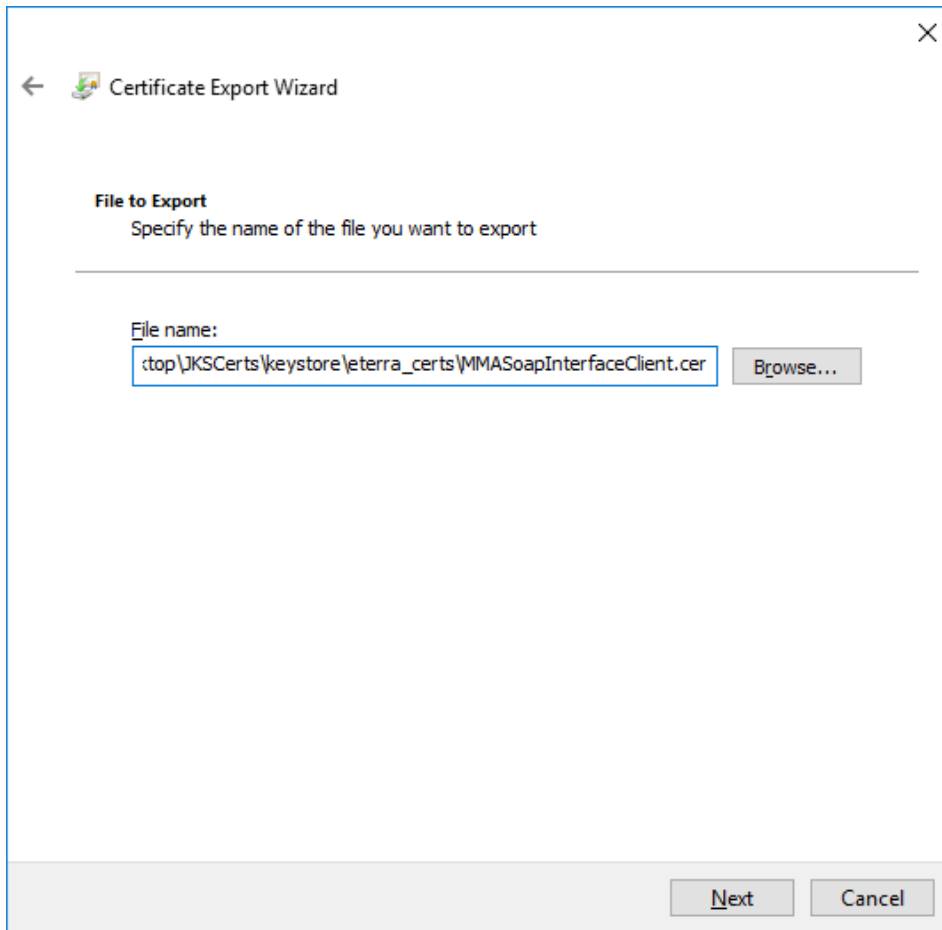
4. At the Export Private Key step, select "No, do not export the private key". Click Next.



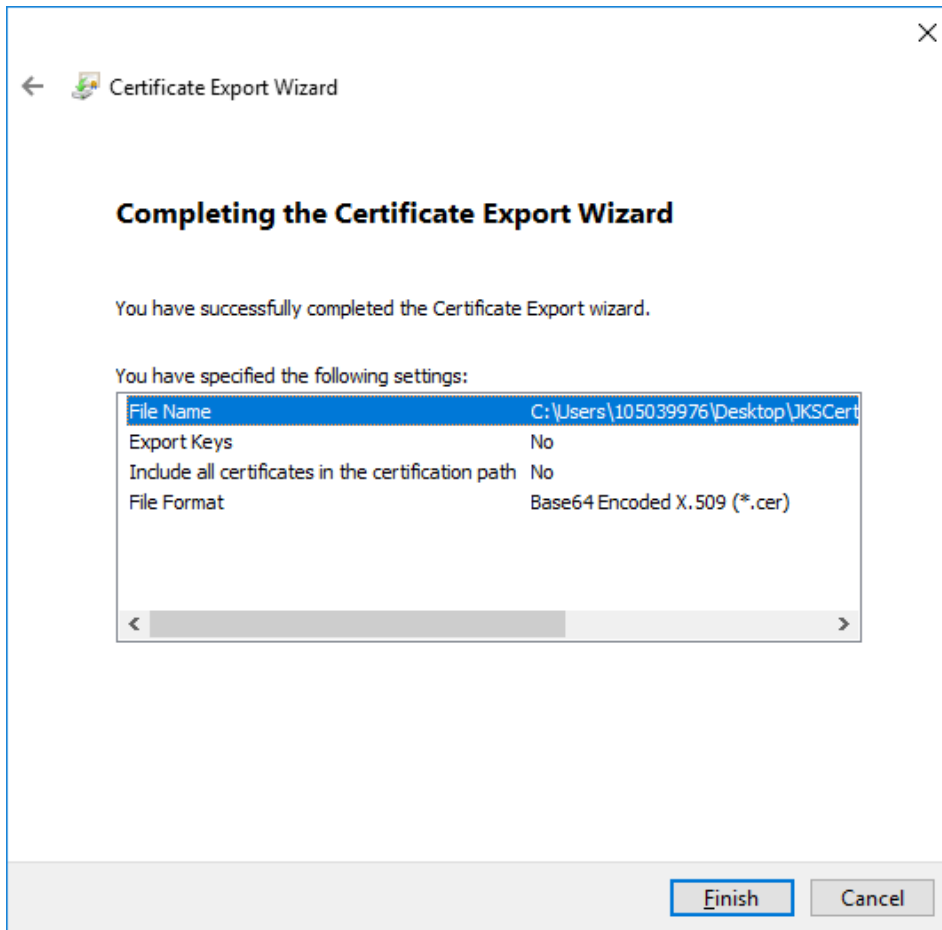
5. At the Export File Format step, select "Base-64 encoded X.509(.CER)". Click Next.



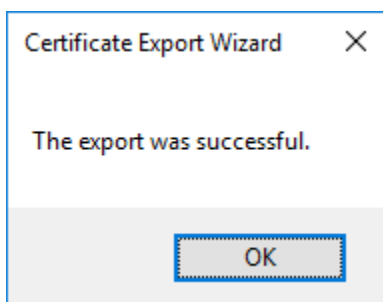
6. At the File to Export step, click Browse and specify the name and location of the file. Click Next.



7. At the Completing the Certificate Export Wizard step, click Finish.



8. A message box appears indicating "The export was successful". Click OK.



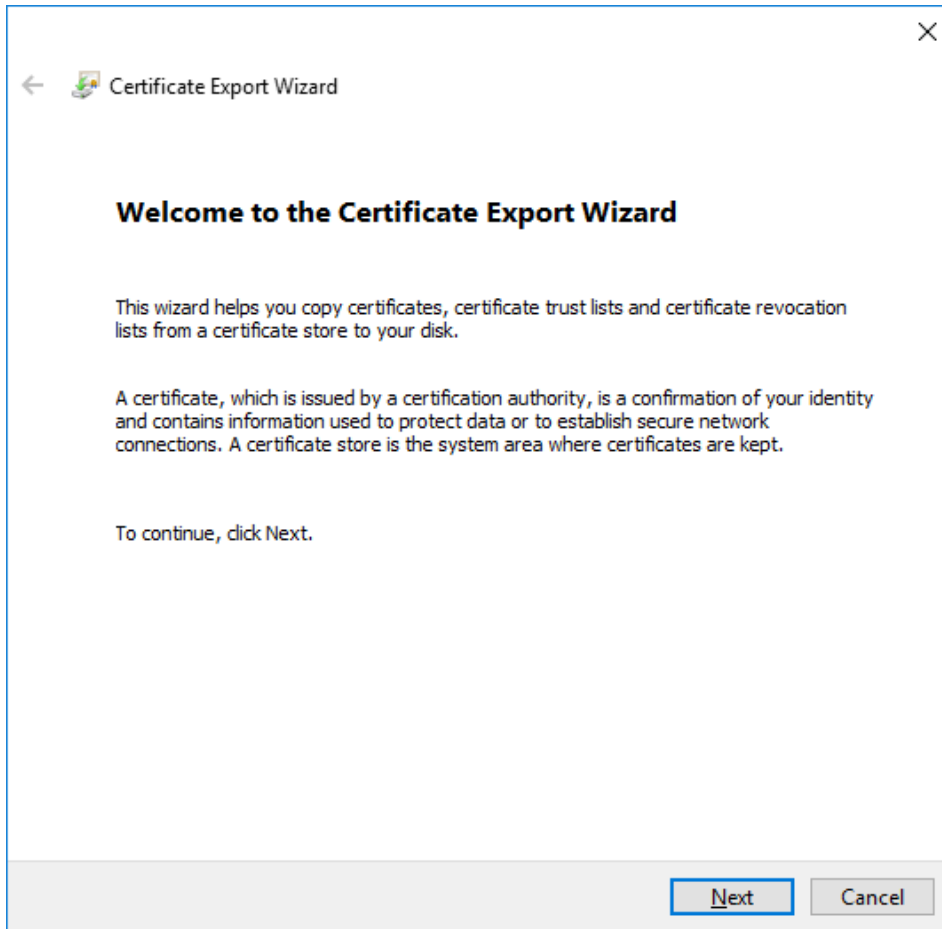
3.2.2. Exporting the MMA Server Certificate for Future Use (Export with Private Keys)

The MMA server certificate needs to be exported along with its private keys so that it can be backed up or used in other MMA servers.

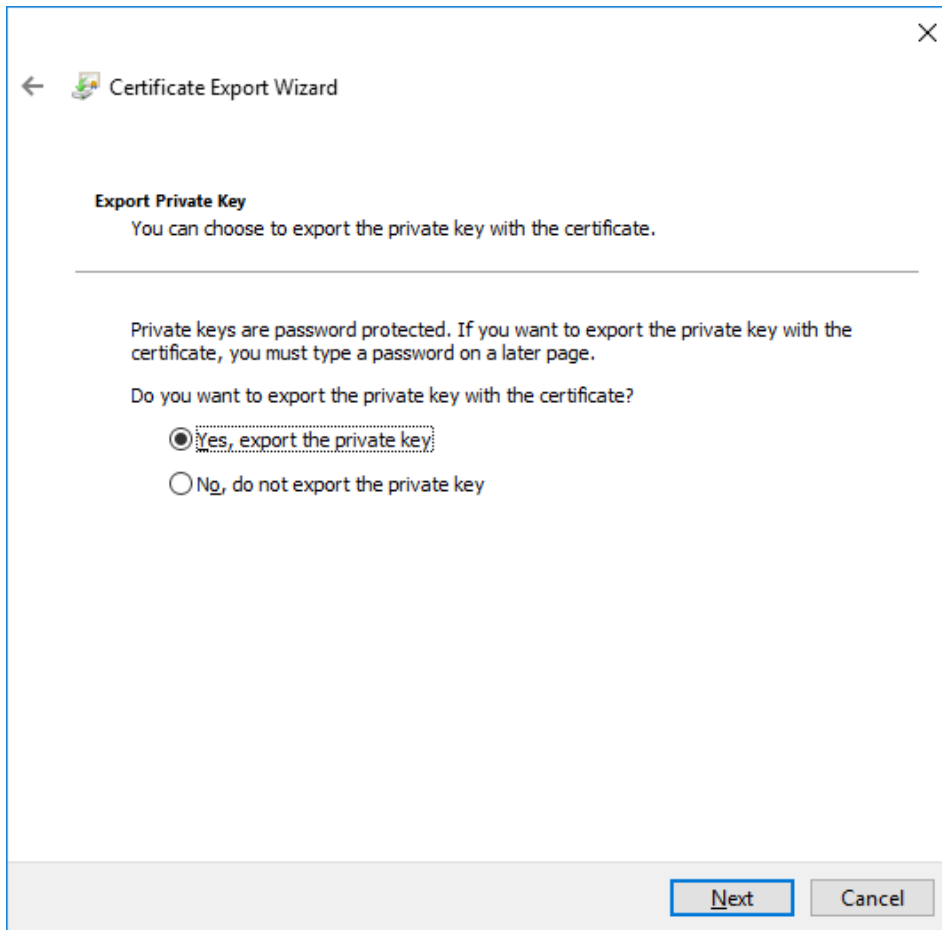
To export a server certificate with private keys:

1. In Microsoft Management Console, select the "MMASoapInterface" certificate from the Personal

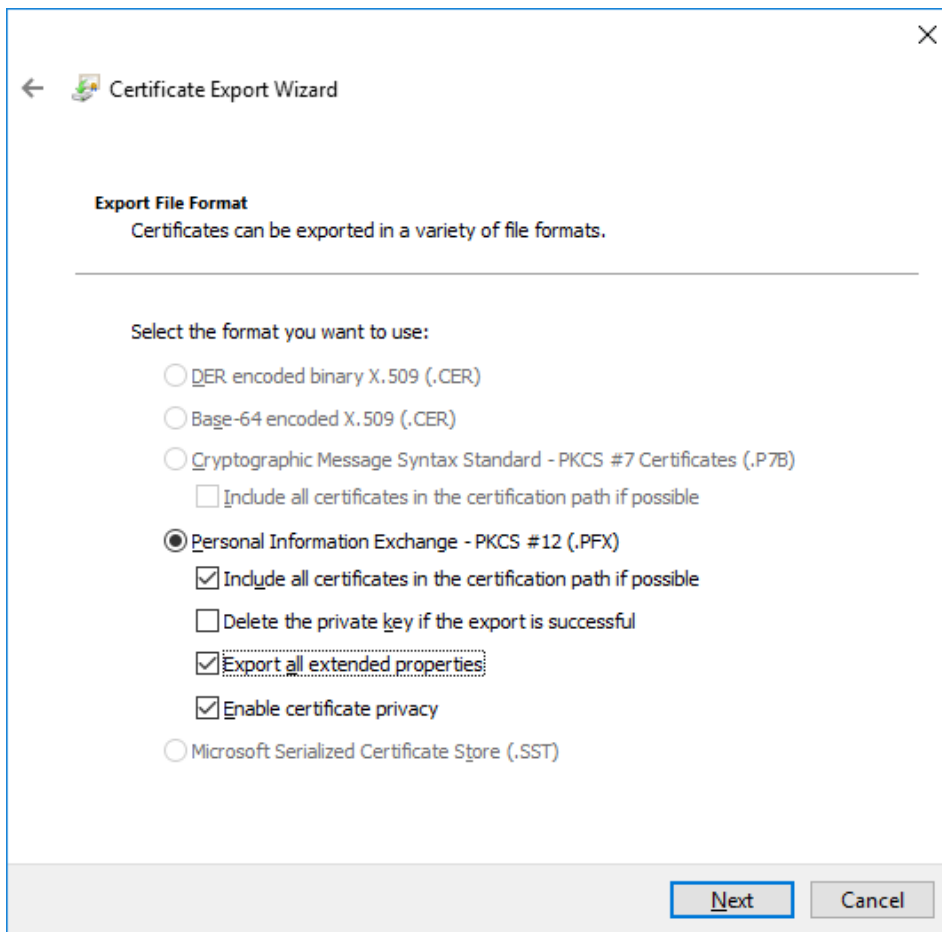
- > Certificates folder.
- 2. Right-click the "MMASoapInterface" certificate, and from the All Tasks menu, click Export.
- 3. At the Certificate Export Wizard window, click Next.



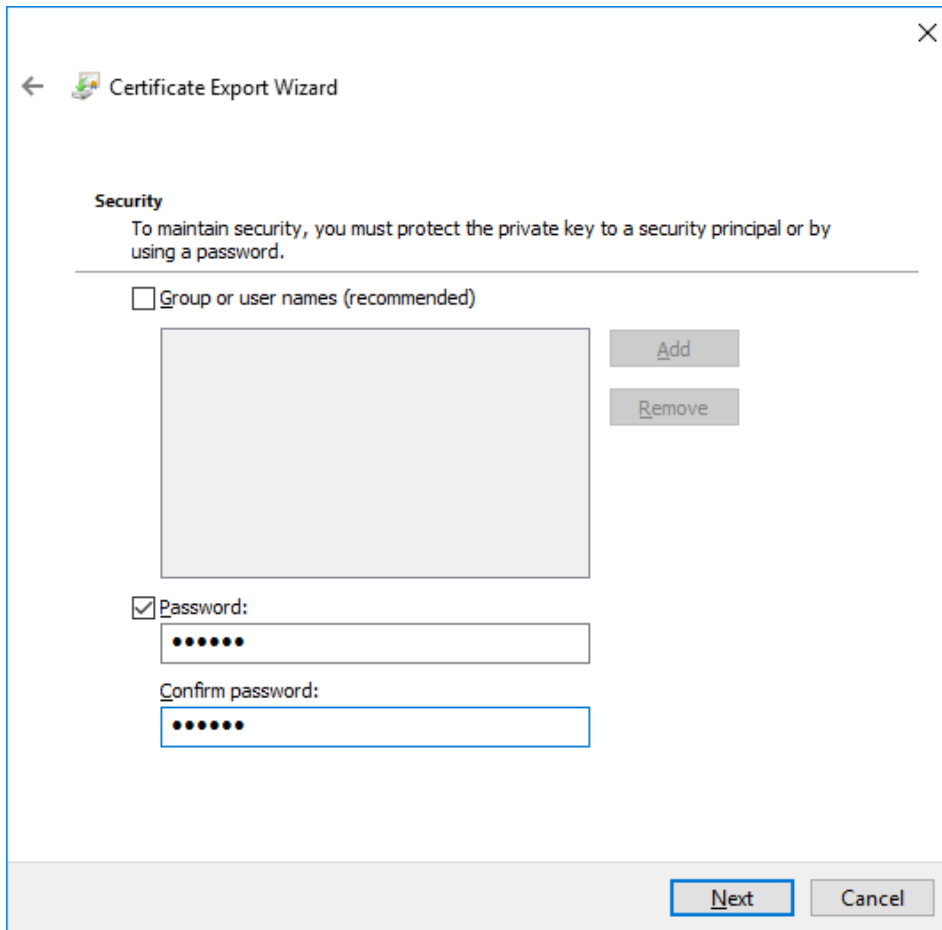
- 4. At the Export Private Key step, select "Yes, export the private key". Click Next.



5. At the Export File format step, select "Personal Information Exchange - PKCS # 12(.PFX)" and select all options except "Delete the private key if the export is successful". Click Next.

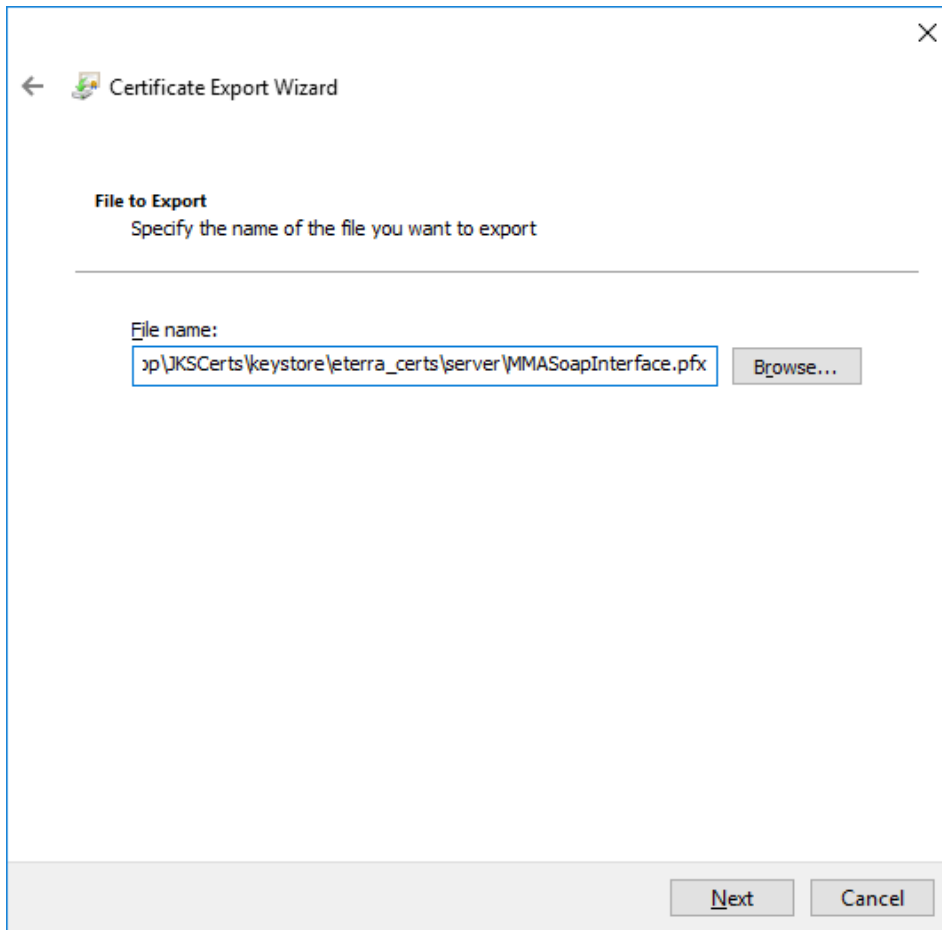


6. At the Security step, select "Password" and enter a password. Click Next.

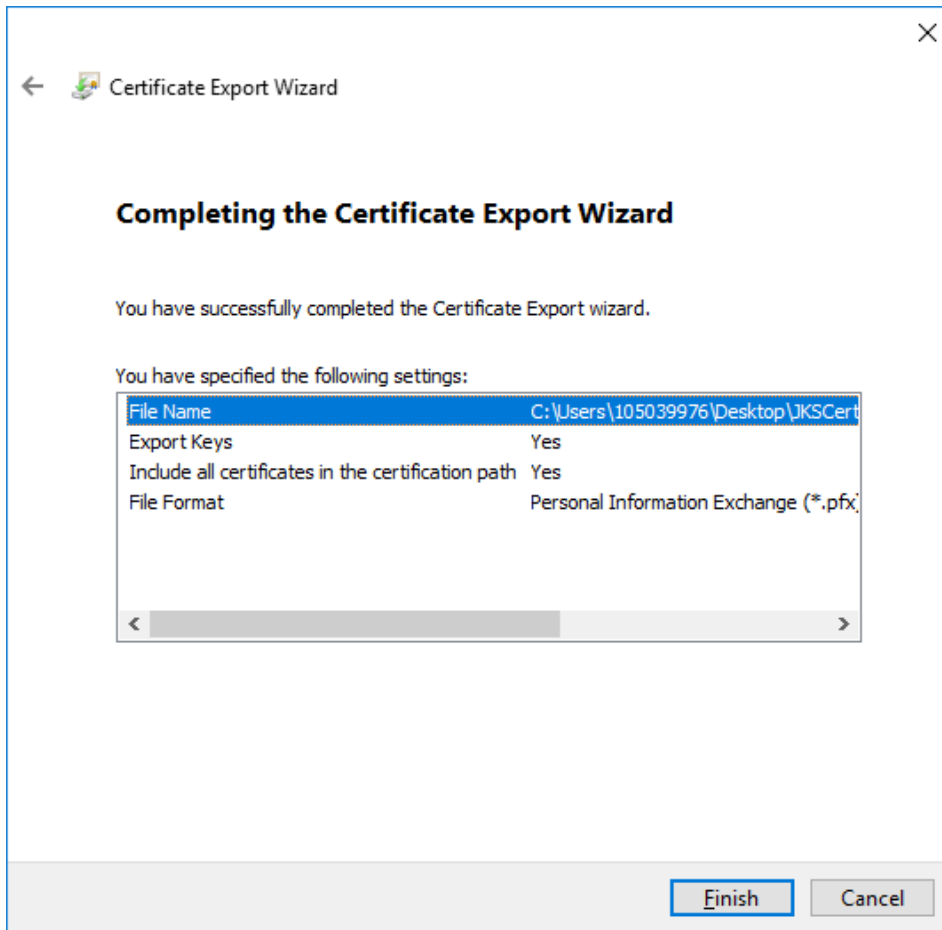


The image shows a Windows-style dialog box titled "Certificate Export Wizard" with a back arrow and a close button (X). The "Security" section contains the instruction: "To maintain security, you must protect the private key to a security principal or by using a password." Below this, there is an unchecked checkbox labeled "Group or user names (recommended)". To its right is a large empty rectangular box, and further right are "Add" and "Remove" buttons. Below the checkbox is a checked checkbox labeled "Password:". This is followed by two text input fields, both containing seven dots to represent masked characters. The first field is for the password, and the second is for the "Confirm password:". At the bottom right of the dialog are "Next" and "Cancel" buttons. The "Next" button is highlighted with a blue border.

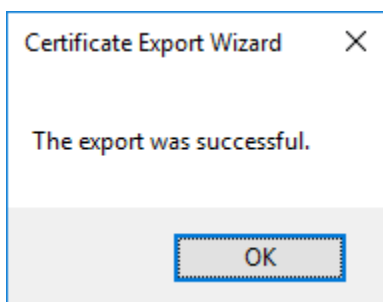
7. At the File to Export step, click Browse and specify the name and location of the file. Click Next.



8. At the Completing the Certificate Export Wizard step, click Finish.



9. A message box appears indicating "The export was successful". Click OK.



3.3. SFTP File Transfer from GISA to Model Manager

In a typical GISA-MMA interface architecture, the Model Manager application runs in a high security operational zone, whereas the Smallworld Electric Office GIS Adapter runs in low security/IT zone.

When GISA publishes substation XML files, they need to be moved to Model Manager folders. To secure the file transfer, use SSH File Transfer Protocol (SFTP). SFTP is a protocol packaged with SSH that works in an equivalent way over a secure connection.

SFTP can leverage a secure connection to transfer files and traverse the filesystem on both the local and remote system. To use SFTP, an SFTP server needs to be installed on the GISA machine (or on the machine where the Model Exchange folder exists). The Model Manager acts as an SFTP client.

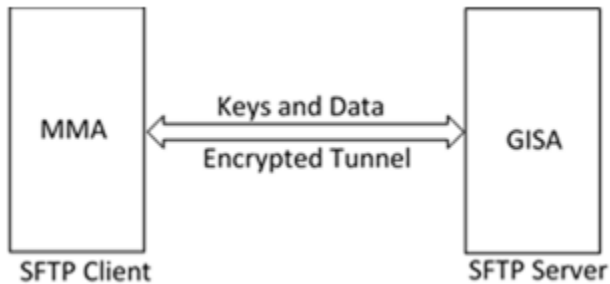


Figure 9. File Transfer among GISA and MMA

When processing a Consumer SOAP message, the Model Manager initiates an SFTP connection to the GISA server and copies the substation zip file to its PreInput folder. While making a connection, the SFTP server authenticates clients that connect to it. Most common methods used for client authentication are password-based authentication and key-based authentication (some servers are configured to require both).

The SFTP server installed on the GISA machine must be configured with key-based authentication. Using key-based authentication, the client stores and uses its own public/private key pair, and the client's public key is stored on the server.

The following sections describe how to configure SFTP on a GISA machine. For more information about setting up SFTP and using the File Updater application, refer to the *File Updater Software Installation and Maintenance Guide*.

3.3.1. Installing VanDyke SecureFX

To set up an SFTP server on the GISA machine (or on the machine where the Model Exchange folder exists), use the third-party software VanDyke VShell. To generate a key pair and test the connection, use the third-party software VanDyke SecureFX.

To install and configure SecureFX on a machine where MMA is running:

1. Run the VanDyke SecureFX installer.

Note

Use VanDyke SecureFX SSH version 6.7 or later.

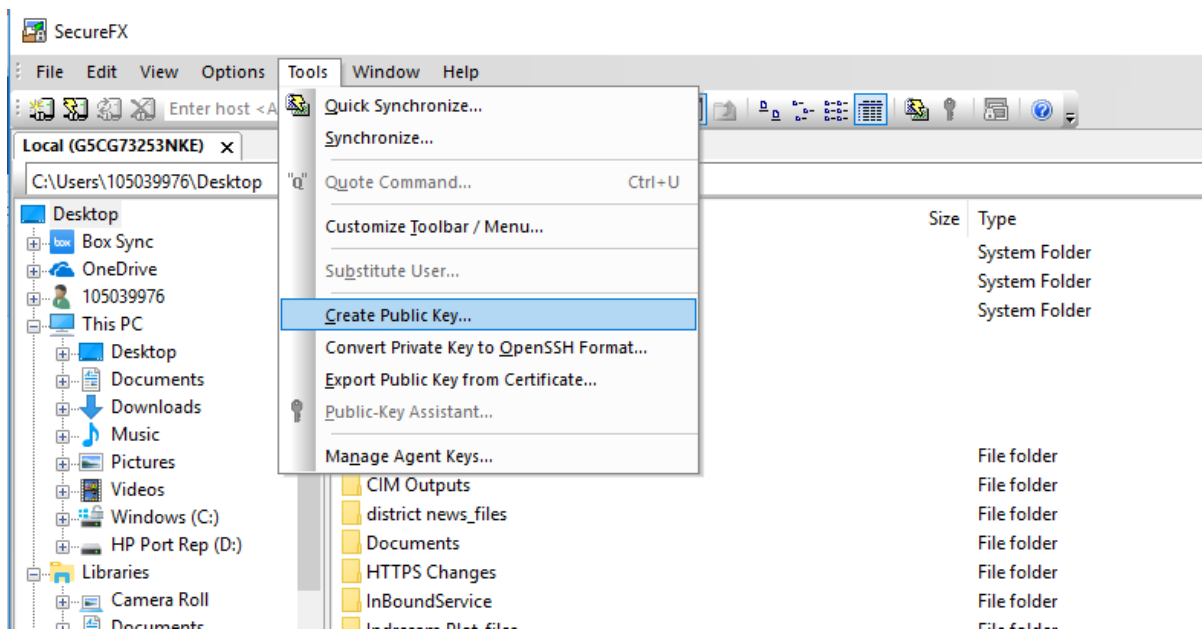
2. During the SecureFX installation, select the Common Profile option.
3. Start SecureFX.
4. When starting SecureFX for the first time, specify a folder where configuration data can be stored. It is recommended that you use the default path.

3.3.1.1. Generating Keys using VanDyke SecureFX

A key pair (private and public key) is generated using VanDyke SecureFX. It is used during client authentication. The public key needs to be stored on the server that uses the SFTP connection.

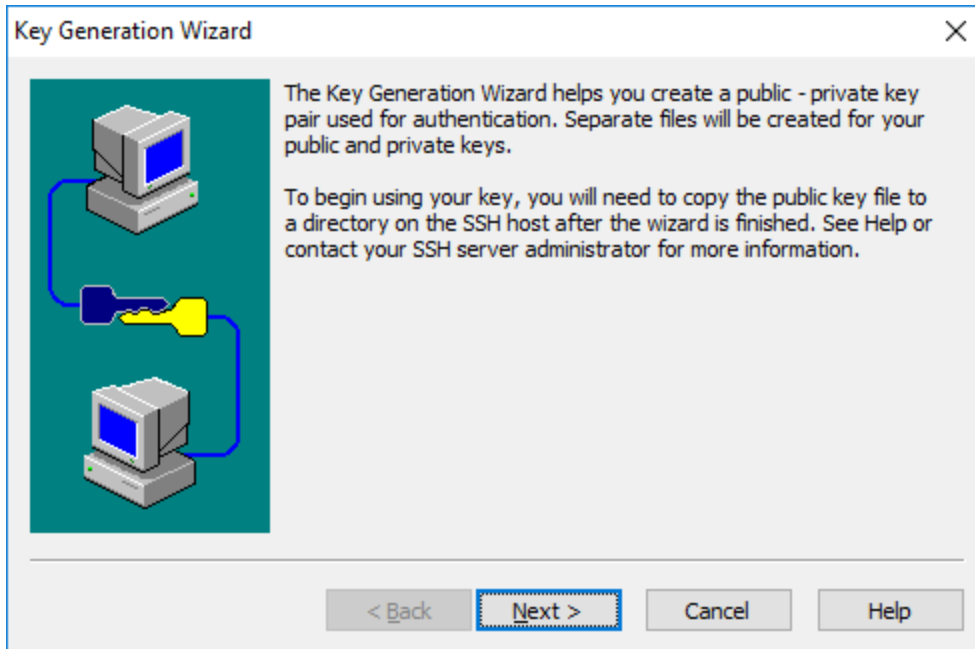
To generate a key pair using SecureFX:

1. Start the VanDyke SecureFX application.
2. From the Tools menu, click Create Public Key.

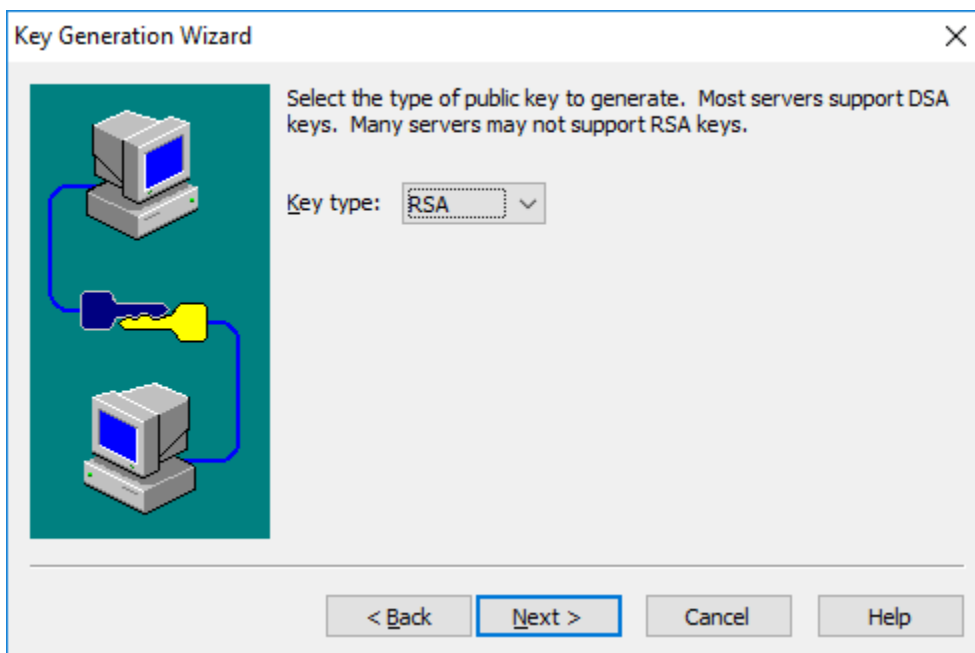


The Key Generation Wizard opens.

3. At the Key Generation Wizard screen, click Next.

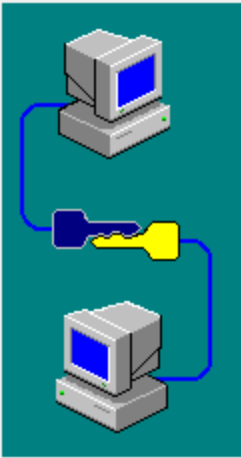


4. Select the RSA key type and click Next.



5. Leave the Passphrase and Comment fields blank and click Next.

Key Generation Wizard [X]



Enter a passphrase which protects your encrypted private key. The passphrase is optional, but if it is not used, the private key will not be encrypted (not recommended).

Passphrase:

Confirm Passphrase:

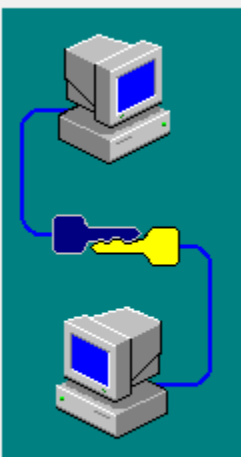
Enter a comment that will be displayed when you are asked for your passphrase. It will be stored with your key.

Comment:

< Back **Next >** Cancel Help

6. Enter 1024 as the Key Length in Bits and click Next.

Key Generation Wizard [X]



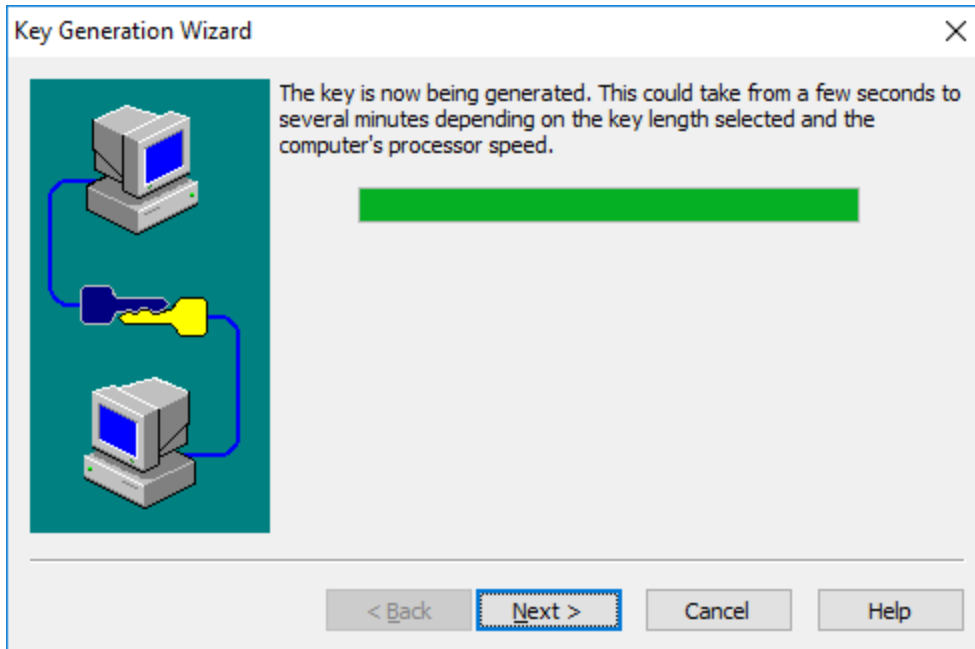
Select the length of your key pair between 512 and 2048 bits.

Key length in bits:

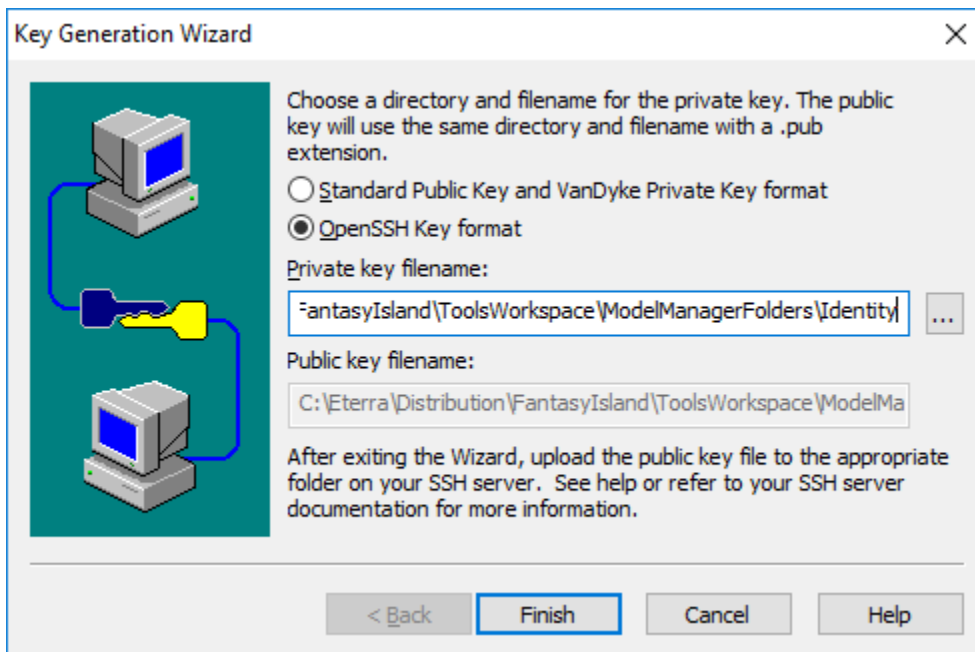
A lower number provides less security, takes less time to generate and authenticates faster. A higher number provides greater security, takes more time to generate, and authenticates more slowly. 1024 is the recommended value.

< Back **Next >** Cancel Help

7. After the key is generated, click Next.



8. Select the OpenSSH Key Format option and then enter the location of the ModelManagerFolders folder. This is the location where the private and public key files will be generated. The public key has a .pub file extension.



9. Click Finish.
10. Use the generated key as a global key.

3.3.2. Installing VanDyke VShell

To Install the VanDyke VShell application:

1. Run the VanDyke VShell installer.

Note

Use VanDyke VShell version 3.8 or later.

Using the default installation settings are recommended.

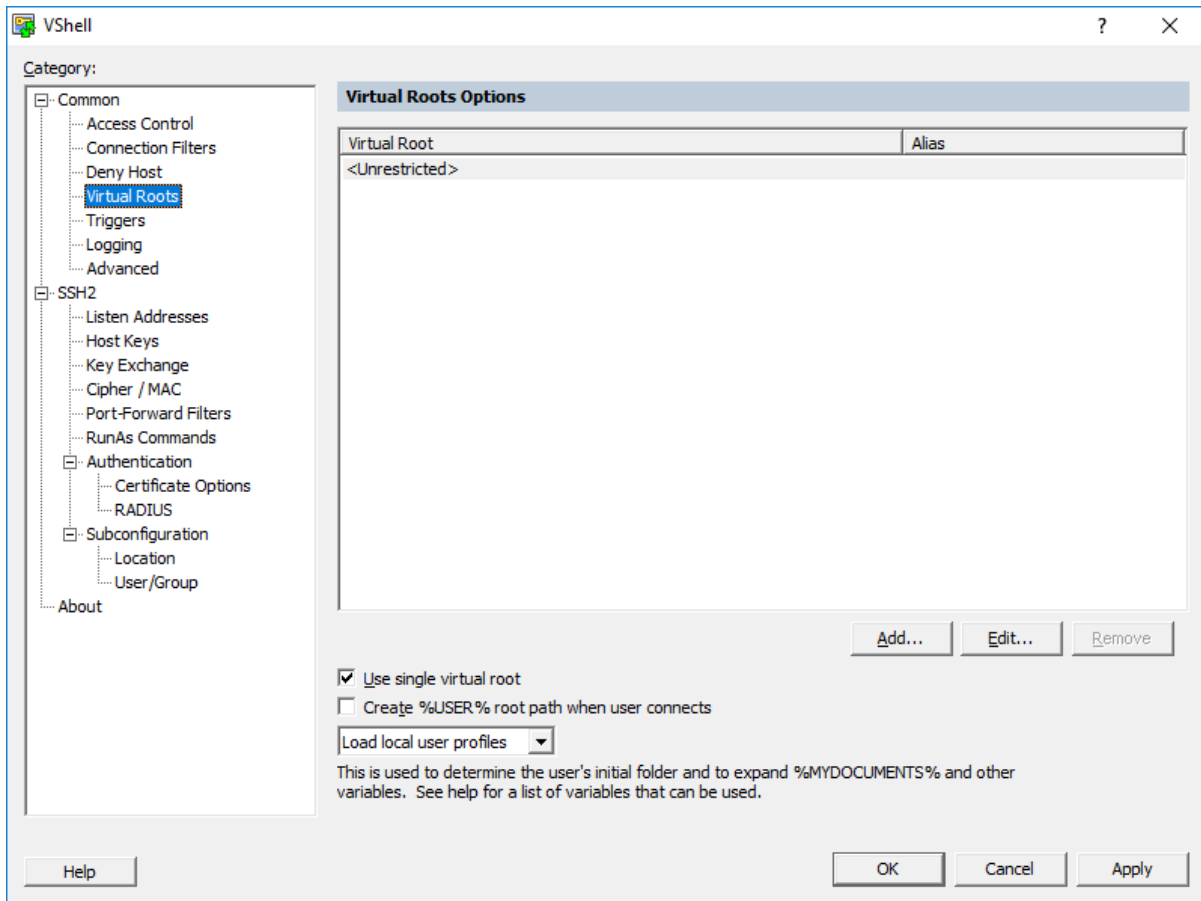
2. After the installation is complete, you need to configure the virtual roots and authentication properties, as described in the following sections.

3.3.2.1. Configuring VShell Virtual Roots (Defining SFTP Folders)

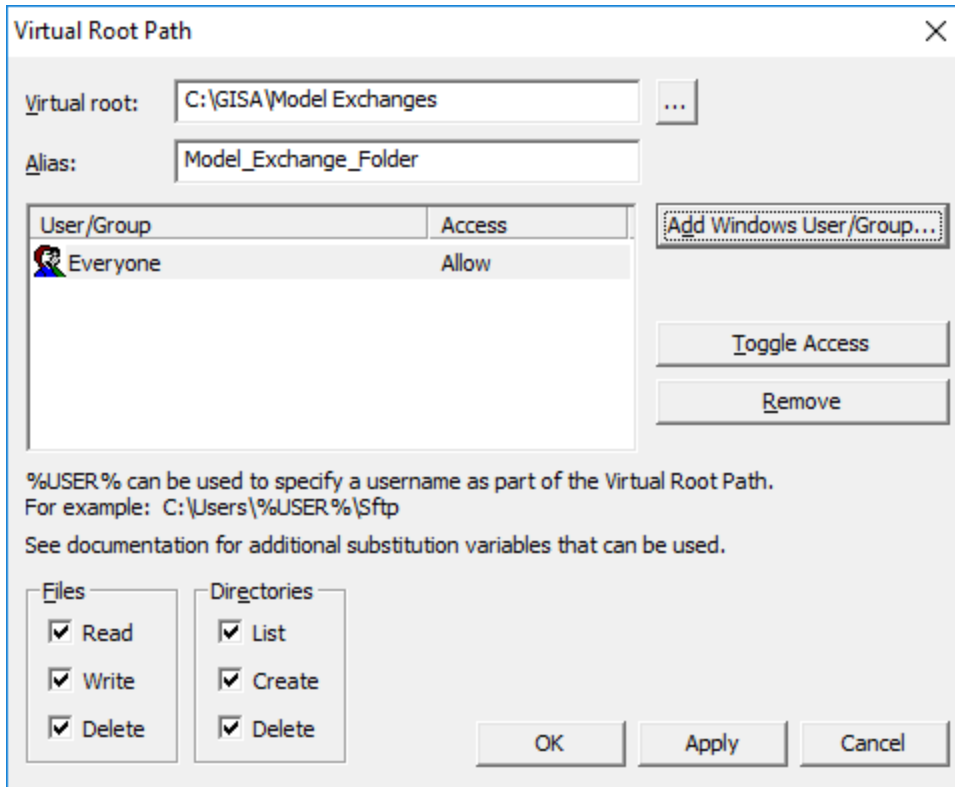
To configure VShell virtual roots:

1. Start the VShell application.
2. Under Category, click Virtual Roots.

The Virtual Roots Options window opens.



3. In the Virtual Root list, select "<Unrestricted>" and click Edit.
4. On the Virtual Root Path form, remove the Everyone user and click OK to save the changes.
5. On the Virtual Roots Options window, click Add.
6. On the Virtual Root Path form, enter the parameters as follows:
 - **Virtual Root:** Set the path to the Model Exchange folder where GISA publishes station DNOM files.
 - **Alias:** Set the alias as appropriate.
 - **User/Group:** Add the Everyone user and grant it permissions to work with files and directories.



7. Click OK to save the changes and close the Virtual Root Path form.
8. Click Apply to save your changes.

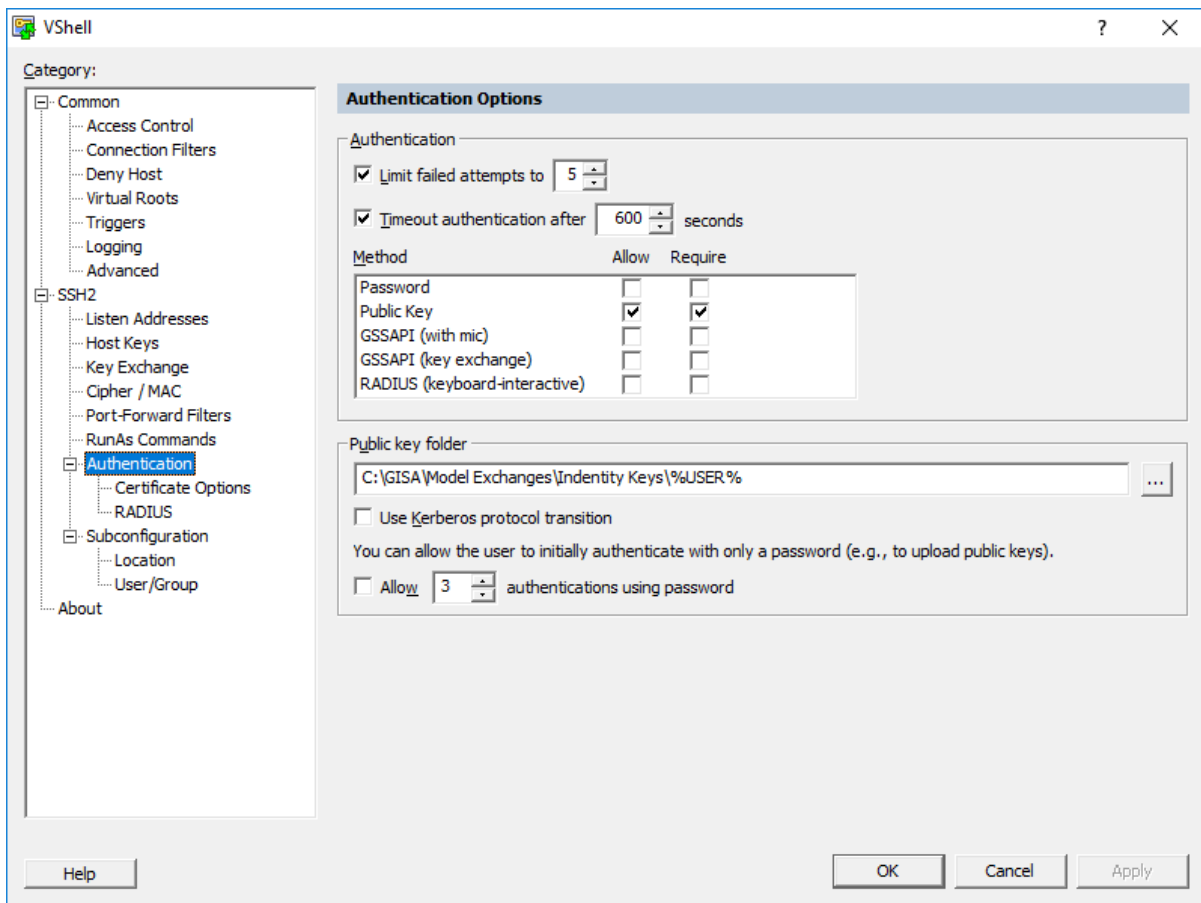
3.3.3. Configuring the VShell Authentication

To configure VShell authentication properties:

1. Open the folder that was added to the virtual root (Model_Exchange_Folder).
2. In the Model_Exchange_Folder folder, create a new folder with the name Identity Keys. This folder is a placeholder for the client's public key.
3. In the Identity Keys folder, create a new folder named after the MMA username (for example, hab06). Place the client's public key file (for example, Identity.pub) that was generated using the VanDyke SecureFX tool into this folder.
4. Start the VShell application.
5. Under Category, click Authentication.

The Authentication Options window opens.

6. Uncheck all authentication methods except Public Key.



7. In the Public key folder group, enter the path to the folder that contains the client's public key (for example, the Identity Keys folder).

Note

The public key folder path format requires the identity key to be placed in the folder with the MMA username.

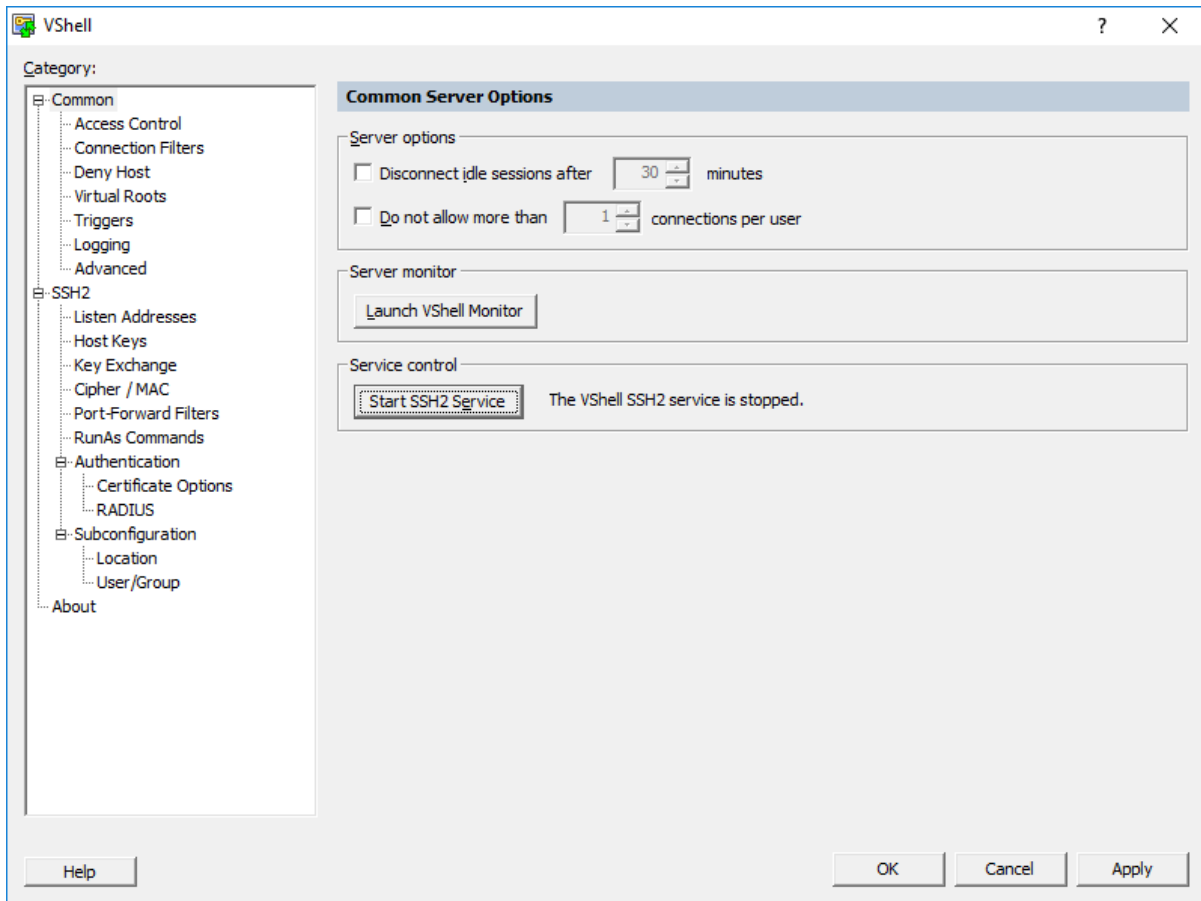
For example, if the MMA username is hab06, and the identity key is contained in the C:\GISA\Model Exchanges\Identity Keys\hab06 directory, the path is C:\GISA\Model Exchanges\Identity Keys\%USER%.

8. Click Apply to save your changes.

3.3.4. Starting the VShell Service

1. Start the VShell application.
2. Under Category, click Common.

The Common Server Options window opens.



3. Click Start SSH2 Service.

The SSH2 Service starts and can be used by GISA and MMA to exchange station DNOM zip files.

3.3.5. Validating the SSH Connection

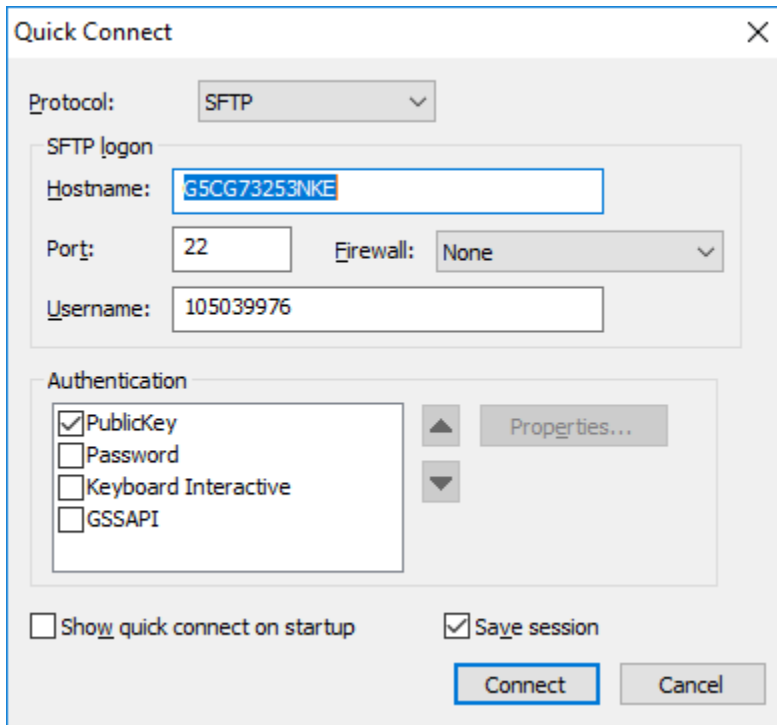
To validate the SSH connection:

1. Start the VanDyke SecureFX application.
2. On the VanDyke SecureFX toolbar, click Quick Connect.

The Quick Connect window opens.

3. Enter the parameters as follows:
 - **Protocol:** Select SFTP.
 - **Hostname:** Enter the computer name, or localhost.
 - **Port:** Use the default value.
 - **Firewall:** None.
 - **Username:** Enter the username.

- **Authentication:** Select PublicKey authentication only.



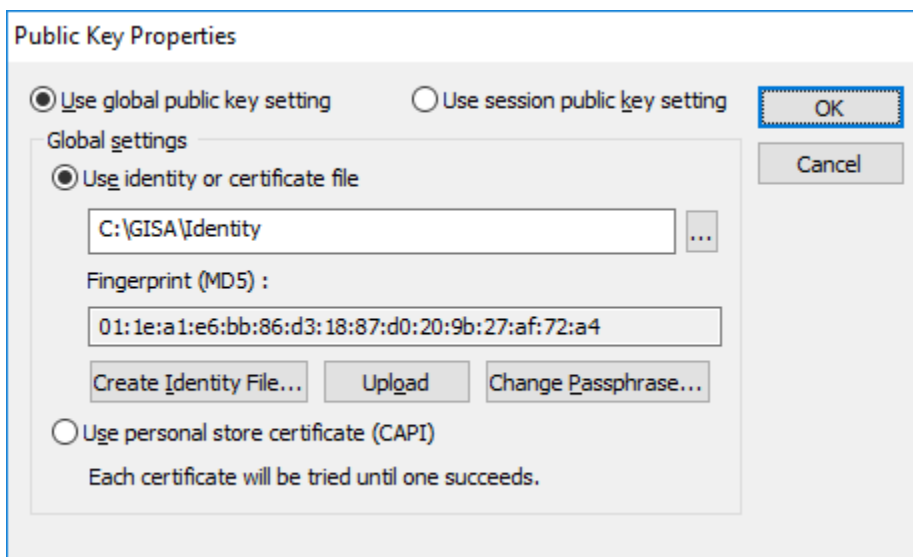
The Quick Connect dialog box is shown with the following settings:

- Protocol:** SFTP
- SFTP logon:**
 - Hostname:** G5CG73253NKE
 - Port:** 22
 - Firewall:** None
 - Username:** 105039976
- Authentication:**
 - ☒ PublicKey
 - ☐ Password
 - ☐ Keyboard Interactive
 - ☐ GSSAPI
- ☐ Show quick connect on startup
- ☒ Save session
- Buttons:** Connect, Cancel, Properties...

- In the Authentication group, after selecting PublicKey, click Properties.

The Public Key Properties window opens.

- Set the path to the previously generated identity key. Use the default settings for the other options.



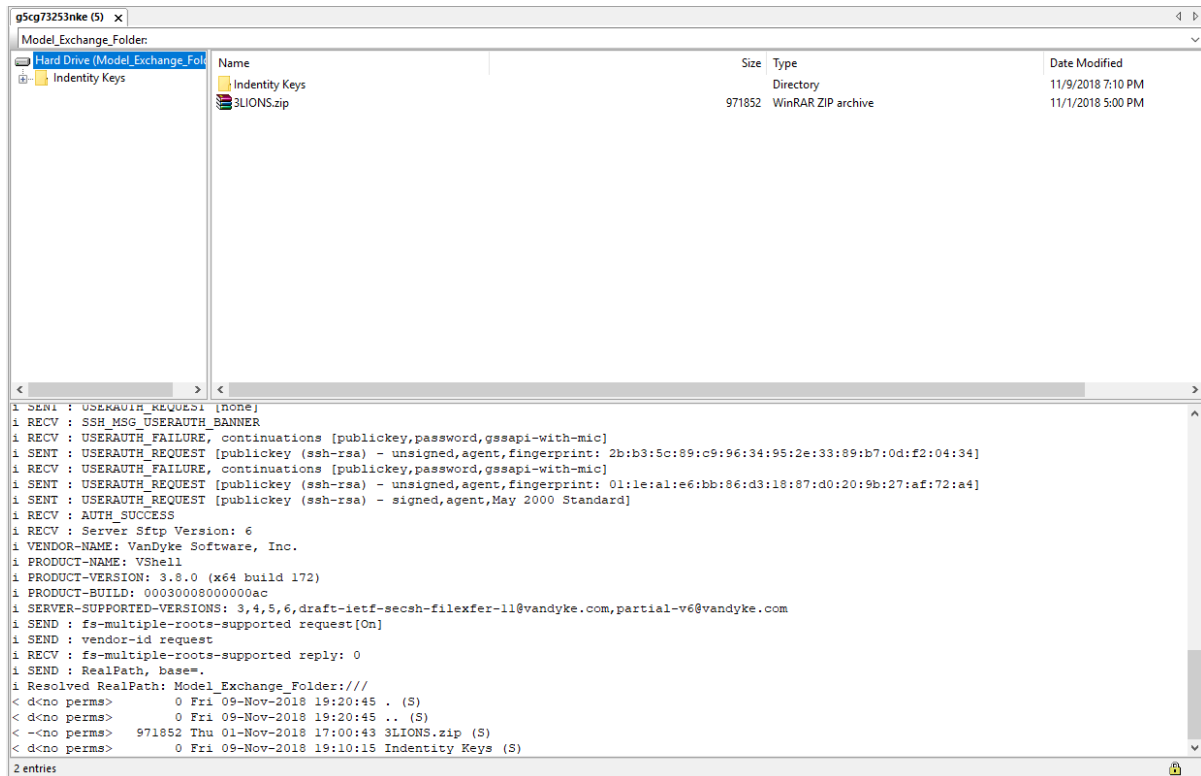
The Public Key Properties dialog box is shown with the following settings:

- Radio buttons:**
 - ☒ Use global public key setting
 - ☐ Use session public key setting
- Global settings:**
 - ☒ Use identity or certificate file
 - Path:** C:\GISA\Identity
 - Fingerprint (MD5):** 01:1e:a1:e6:bb:86:d3:18:87:d0:20:9b:27:af:72:a4
 - Buttons:** Create Identity File..., Upload, Change Passphrase...
 - ☐ Use personal store certificate (CAPI)
- Buttons:** OK, Cancel

- Click OK to save your changes and close the Public Key Properties display.
- On the Quick Connect display, click Connect to save your changes and establish the SFTP

connection.

You are now connected to the SFTP server without being asked for login and password credentials. Here, you can check the defined SFTP folders (virtual roots).



4. Configuring HTTPS for the REST Web Service

This chapter describes how to enable HTTP Transport Layer Security (TLS) for the Model Manager REST Interface Server and Client to accept and send REST messages to external systems over a secure channel.

REST messages are exchanged between Model Manager Application (MMA) and External Systems over HTTPS. External systems could be any GIS systems, planning applications, or other applications within the utility. Example of external systems are Yambay, CYME, Short Term Forecast Engine (STFE), Optimization Engine (OE), and Analytics Applications. MMA can also process station external files published by external applications.

To provide the security for REST messages exchanged, HTTP Transport Layer Security (TLS) with mutual certificate authentication is used. TLS is a security protocol that is based on the Secure Sockets Layer (SSL) protocol. TLS secures network protocol which is used by the web service for communication. TLS uses digital certificates to authenticate the user as well as authenticate the network.

4.1. Creating Certificates

Certificates are required to enable HTTPS on the web server and to enable client authentication.

For production, it is recommended to use certificates issued by a certificate authority. For testing and demo purposes, you can use self-signed certificates.

This section describes how to create certificate authority (CA) signed certificates using the CreateTestCertificates.py Python script. This script uses Be-Your-Own-Certificate-Authority (BYOCA) as a certificate authority. The script can create Root Certificate Authorities, Host Certificates, and Certificate Chains. You can create certificates for multiple hosts, which are specified by their Fully Qualified Domain Names (FQDNs) in the first parameter. You can adjust other parameters to mimic expected production scenarios. This script internally uses Python's Cryptography and idna libraries. This script creates two sets of certificate chains: one set is used for MMA and another set is used for the external systems.

Only non-encrypted PKCS#8 PEM Key files and PEM Certificates are currently supported and created using this script. Therefore, after generating the certificates, use the SSL exe to convert a PEM certificate and the certificate's private key to a Personal Key Store (PKCS12) certificate.

Use the following command for converting a certificate from one type to another. Enter a passphrase while exporting the certificate. Same passphrase will be used further in the document.

```
"<open SSL exe path>" pkcs12 -export -in <certificate file name> -inkey <key file name> -name  
<friendly name> -out <PKCS certificate file name>
```

For example:

```
"C:\Eterra Distribution\MMAGISASecurity\OpenSSL\bin\openssl.exe" pkcs12 -export -in localhost.crt
```

```
-inkey localhost.key -name MMARESTInterface -out Certificate-PKCS-12.p12
```

4.2. Accepting REST Messages from External Systems

To accept REST messages from external systems (REST clients) over a secure channel, the self-hosted MMA RESTful web service is configured with an SSL (X.509) certificate. This allows the REST client to verify the identity of the MMA server. The REST client is also configured with an X.509 certificate, which allows the MMA RESTful web service to verify the identity (client authentication) of the REST client. The MMA server's certificate must be trusted by the REST client and the REST client's certificate must be trusted by the MMA server.

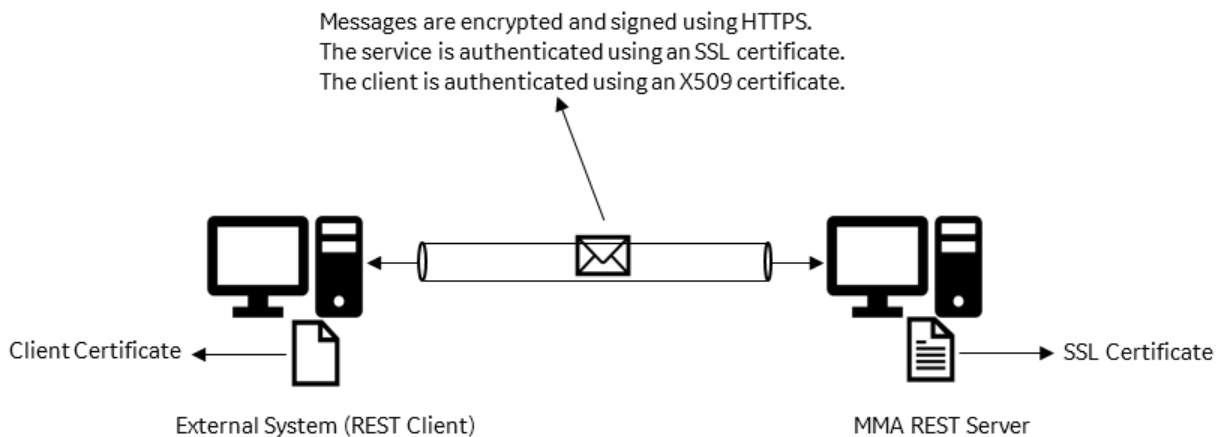
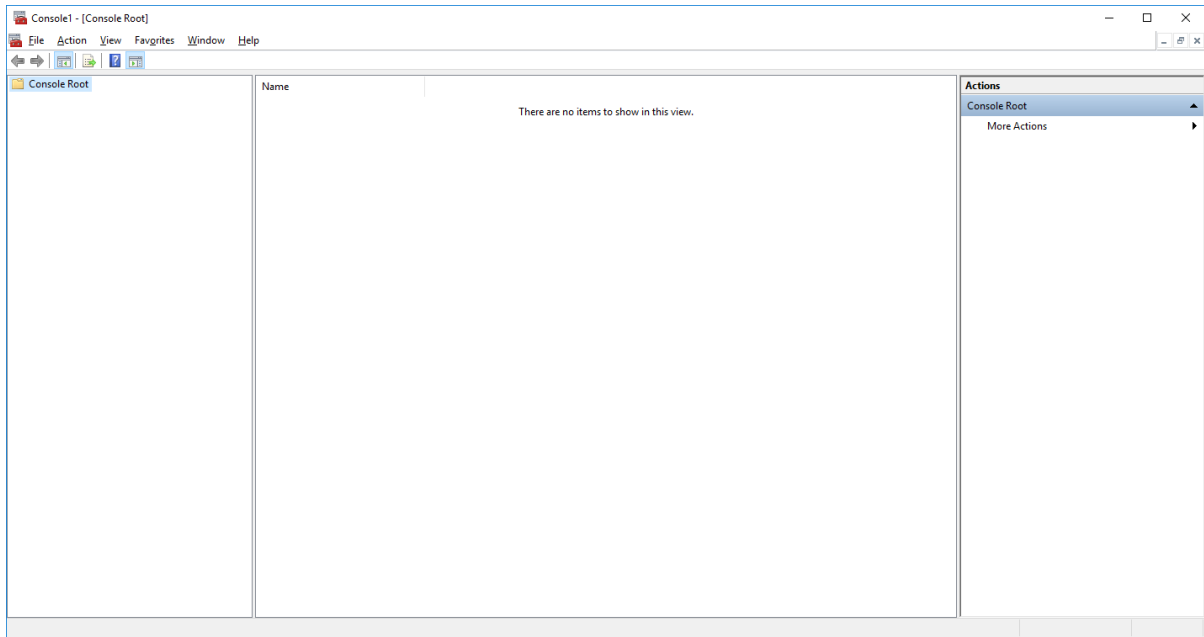


Figure 10. MMA REST Server and External System Interfacing using TLS with Certificate Authentication

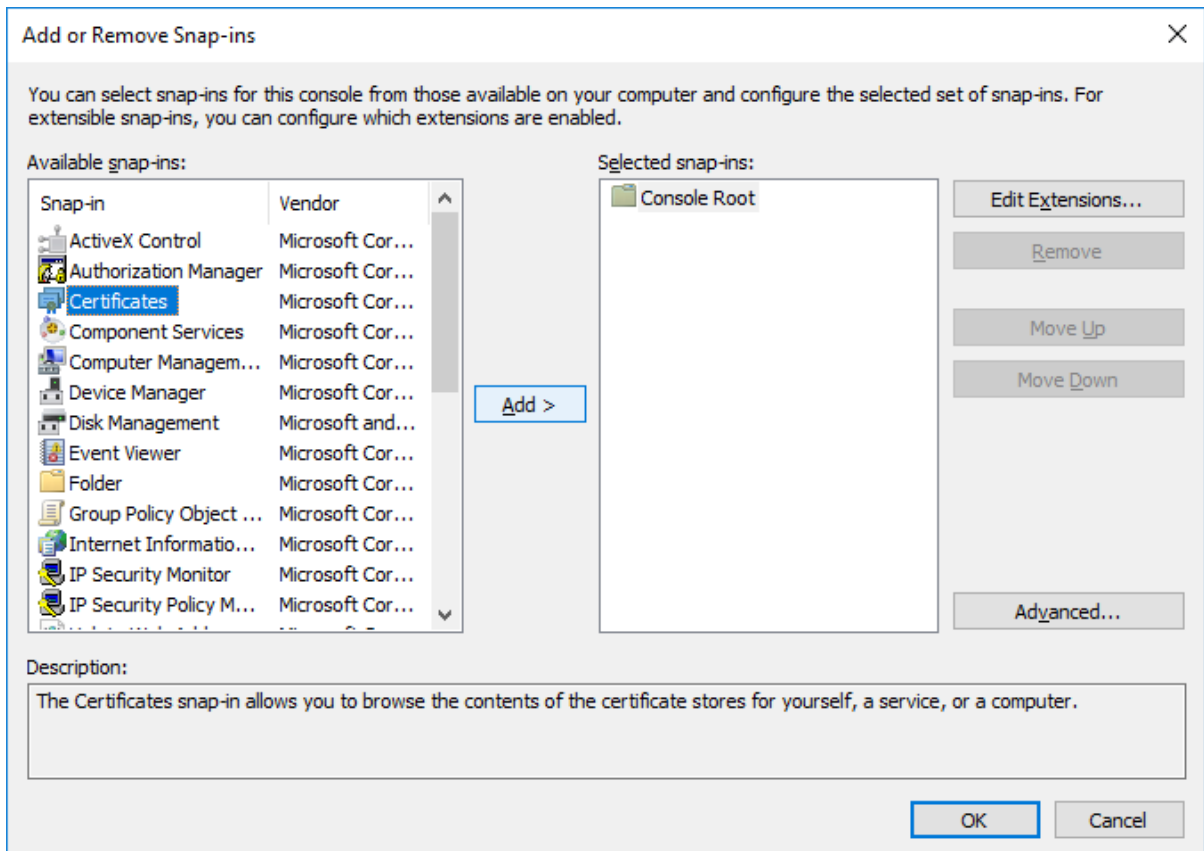
4.2.1. Configuring MMA Server Certificate

To configure the certificate to be used for the MMA REST web service:

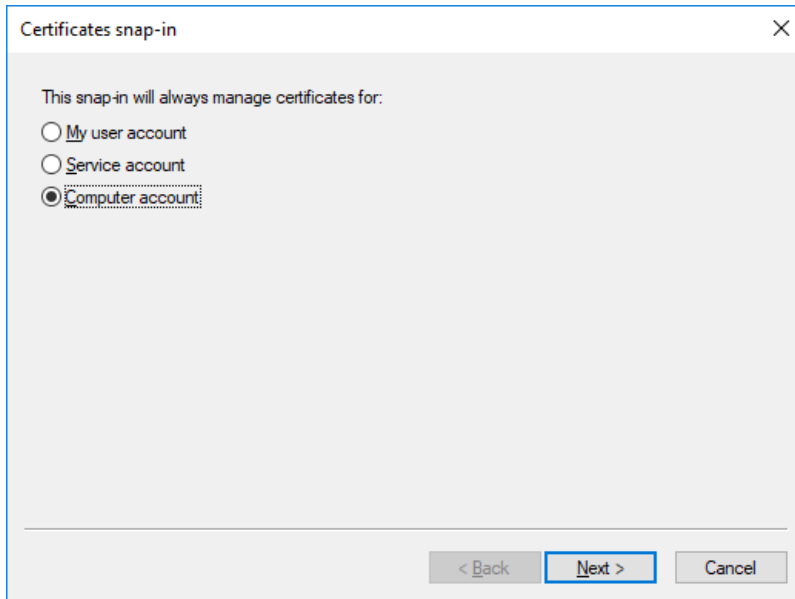
1. Open the Microsoft Management Console application (for example, type `mmc` in the Windows Start menu).
2. Click File, then click Add/Remove Snap-in.



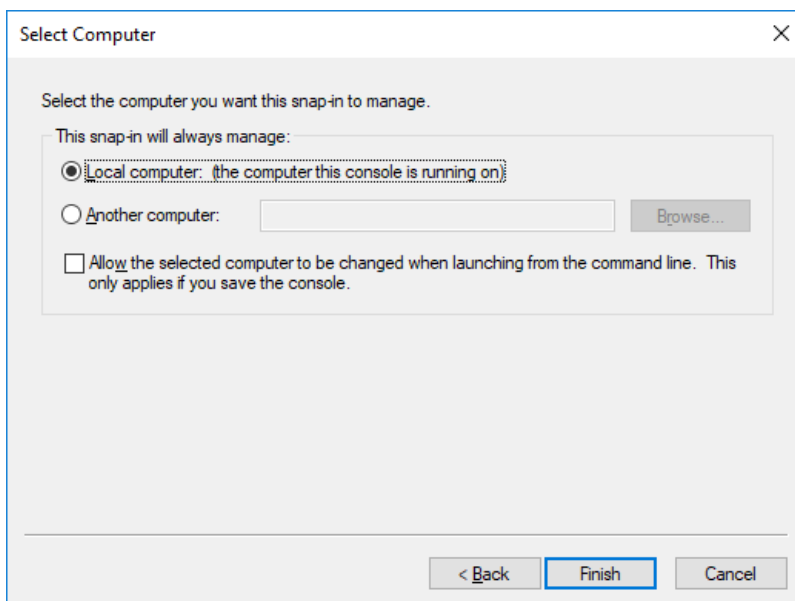
3. Select Certificates from the available snap-ins, and then click Add.



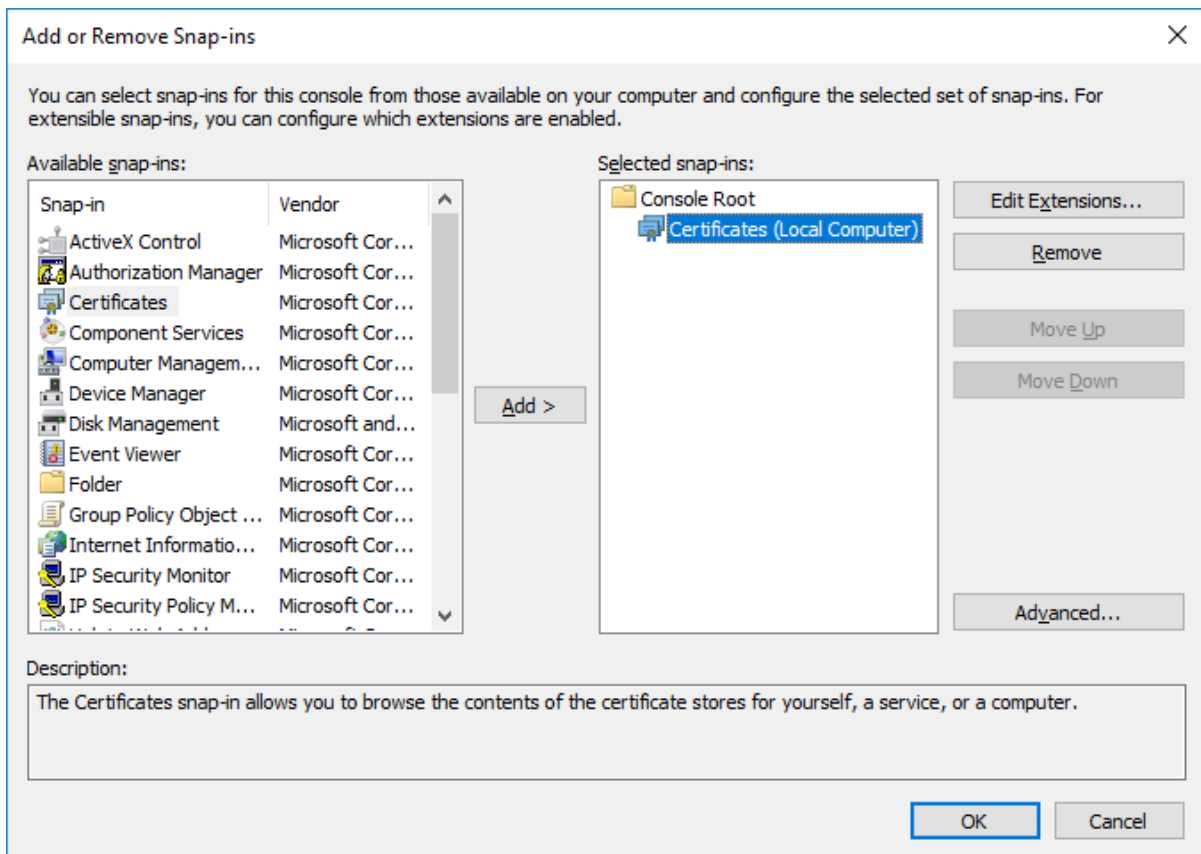
4. Select "Computer account" and click Next.



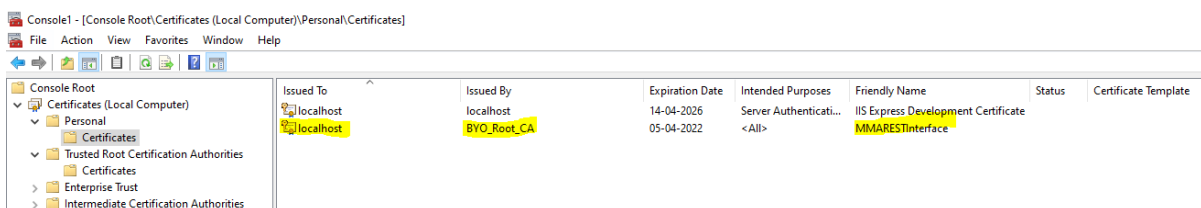
5. Select "Local Computer: (the computer this console is running on)" and click Finish.



The Certificates (Local Computer) entry is added to Console Root in the Selected Snap-Ins panel.



6. Click OK.
7. From the Console Root, click **Personal > Certificates > All Tasks > Import**, and then select the `cert/MmaRestServer.p12` certificate. This imports the certificate to the local computer.
8. After importing, right-click the certificate, open the certificate properties, open the General tab, and change the Friendly Name parameter to `MMARESTInterface`.



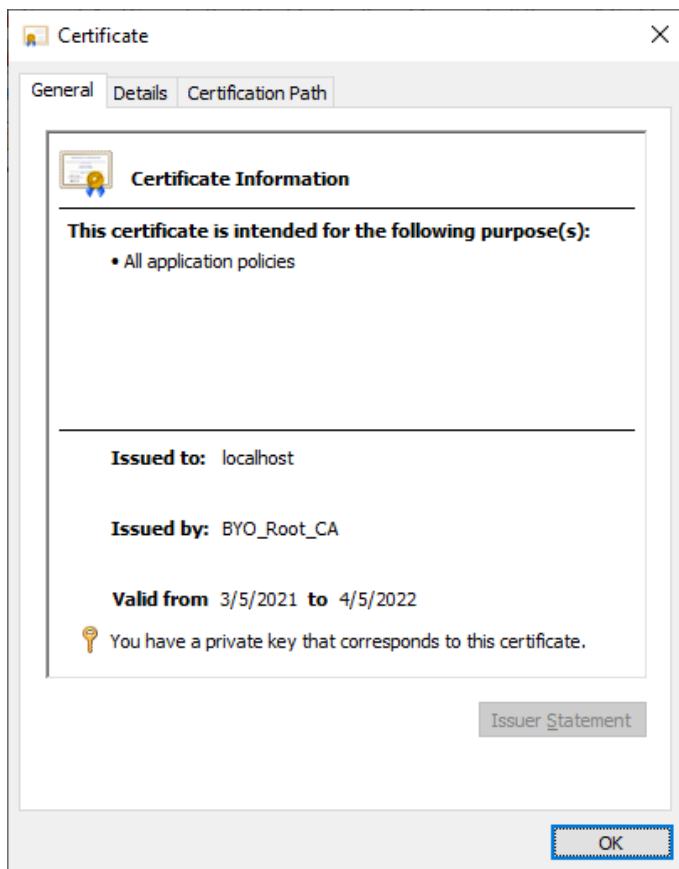
9. From the Console Root, click **Trusted Root Certification Authorities > Certificates > All Tasks > Import**, and then select the `ca/BYO_Root_CA.crt` certificate. This imports the certificate authority to Trusted Root Certification Authorities.
10. After importing, right-click the certificate, open the certificate properties, open the General tab, and change the Friendly Name parameter to `MMARESTInterfaceCA`.

Console1 - [Console Root\Certificates (Local Computer)\Trusted Root Certification Authorities\Certificates]

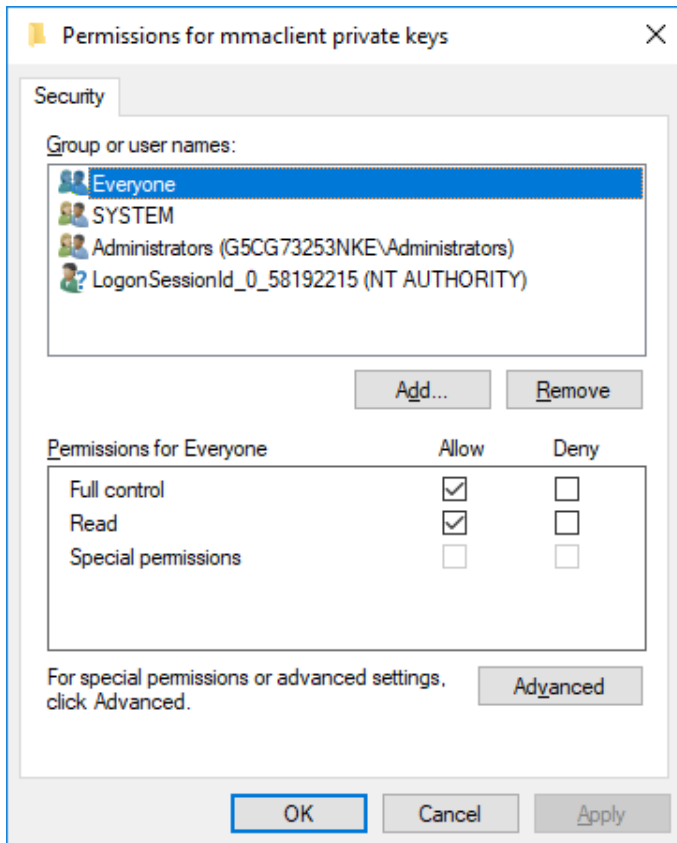
File Action View Favorites Window Help

Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name
AAA Certificate Services	AAA Certificate Services	01-01-2029	Client Authenticati...	Sectigo (AAA)
AddTrust External CA Root	AddTrust External CA Root	30-05-2020	Client Authenticati...	Sectigo (AddTrust)
Assistant Root CA	Assistant Root CA	07-08-2023	<All>	<None>
Baltimore CyberTrust Root	Baltimore CyberTrust Root	13-05-2025	Client Authenticati...	DigiCert Baltimore Root
BYO_Root_CA	BYO_Root_CA	05-04-2022	<All>	MMARESTInterfaceCA
Certum CA	Certum CA	11-06-2027	Client Authenticati...	Certum
Check MyTech SN	Check MyTech SN	26-04-2031	<All>	<None>
Class 3 Public Primary Certificat...	Class 3 Public Primary Certificatio...	02-08-2028	Client Authenticati...	VeriSign Class 3 Public Primary CA

After importing the CA certificate, the MMARESTInterface certificate's General tab looks as follows:



11. From the MMARESTInterface certificate right-click menu, select All Tasks > Manage Private Keys.
12. Add permissions to applicable Windows users for using the certificate private keys.



4.2.2. Adding SSL Binding

HTTP transport layer security requires an SSL (Secure Sockets Layer) certificate to bind with a port where a web service is running.

To bind an SSL certificate to a port used by MMA RESTful web service:

1. Open a command prompt as Administrator.
2. Execute the following command to remove the existing binding of the port with any other SSL certificate. If no bindings exist, you will get the "SSL Certificate deletion failed" error.

```
netsh http delete sslcert ipport=0.0.0.0:<#port number>
```

For example:

```
netsh http delete sslcert ipport=0.0.0.0:9000
```

3. Execute the following command to add a binding for the certificate with the port.

```
netsh http add sslcert ipport=0.0.0.0:<#port number> certstorename=MY certhash=<#Certificate Thumbprint> appid=<#Application GUID>
```


For example:

```
netsh http add sslcert ipport=0.0.0.0:9000 certstorename=MY  
certhash=4b18e2c85b55718dbc18037f8bf8d4c84b590d6 appid=\{C60EF341-984E-4262-AE0F-8DEBB6D98255}
```

Where:

- **ipport:** The IP address and port for the binding.
- **certstorename:** The store name for the certificate. Required for Hostname-based configurations. Defaults to MY for IP based configurations. The certificate must be stored in the local machine context.
- **certhash:** The SHA hash of the certificate. This hash is 20 bytes long and specified as a hex string. To get the cert hash, double-click the MMARESTInterface certificate in the certificates list, and then copy the Thumbprint hexadecimal string on the Details tab. Then, paste the value in a text editor and remove the first character (?) and any empty spaces.
- **appid:** The GUID to identify the owning application. GUID of the Model Manager Application is C60EF341-984E-4262-AE0F-8DEBB6D98255.

Note

You must use the same port number in the MMA Configuration > RESTful Interface > RESTful Interface Server > RESTful Web Service End Point Address configuration in the Model Manager Configuration Editor.

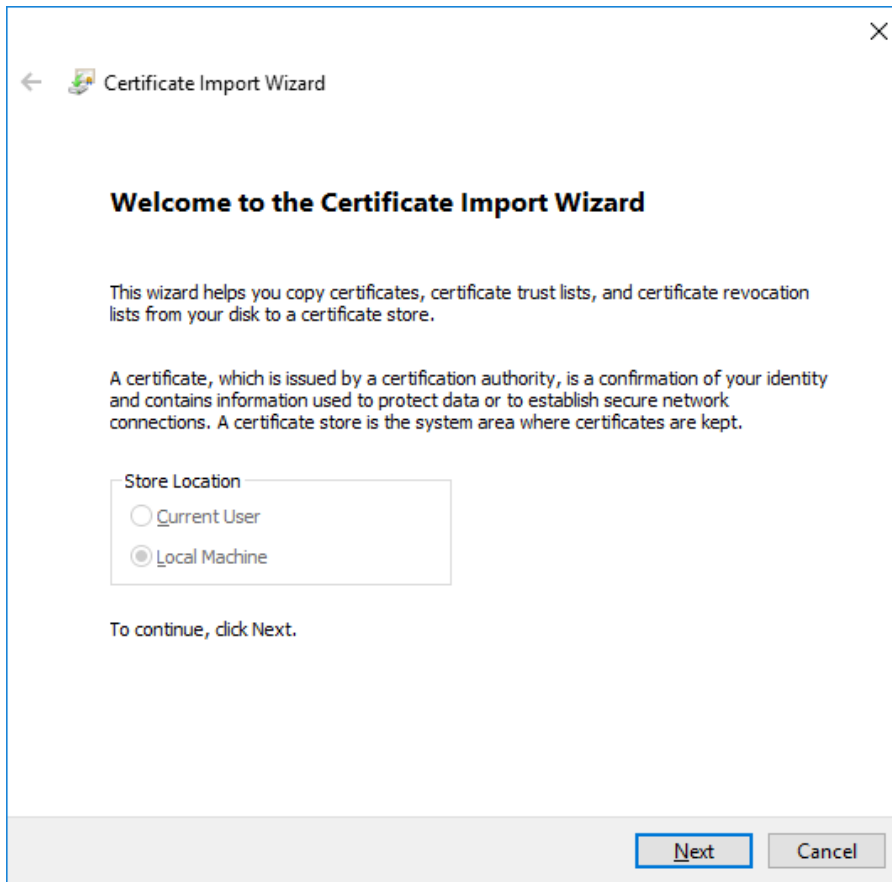
4.2.3. Importing External System CA Certificate to Trusted Root CA Folder

In client certificate authentication, clients are authenticated without a user login and password. Instead, the server asks the client to show its certificate and verifies its identity. During verification, the server checks whether the client certificate chains to a trusted root certificate authority.

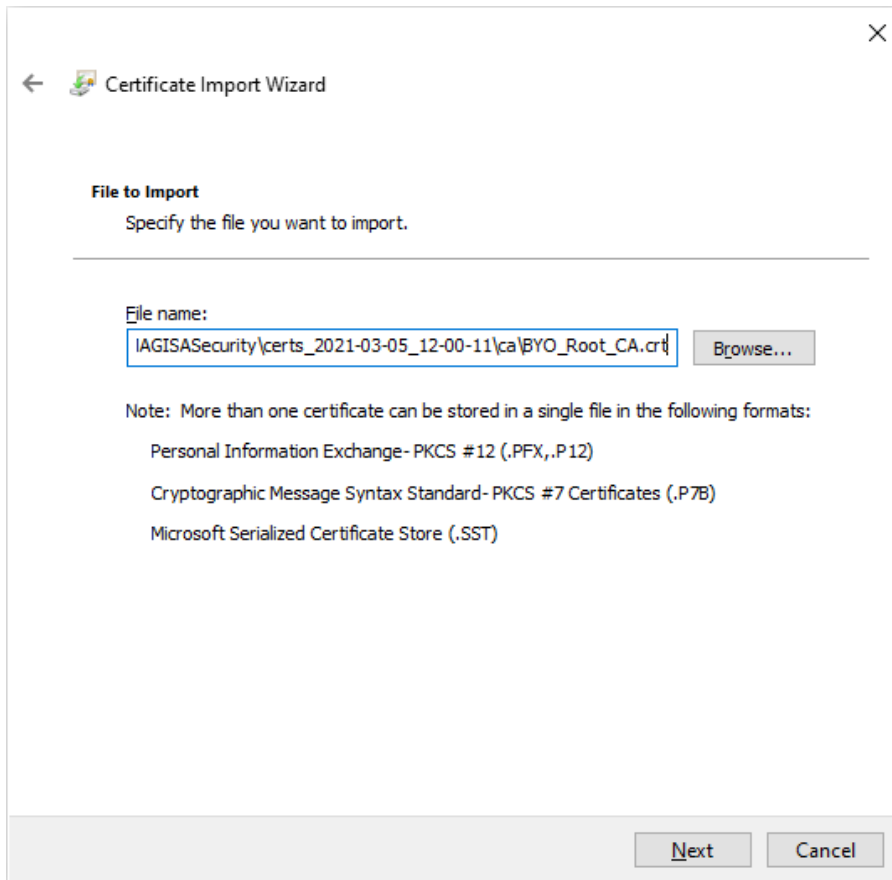
MMA REST Interface server checks the identity of an external system (REST client). For client authentication to be successful, a certificate signed by the External System CA must be imported to the Trusted Root CA folder.

To import a certificate signed by the External System CA to the Trusted Root CA folder:

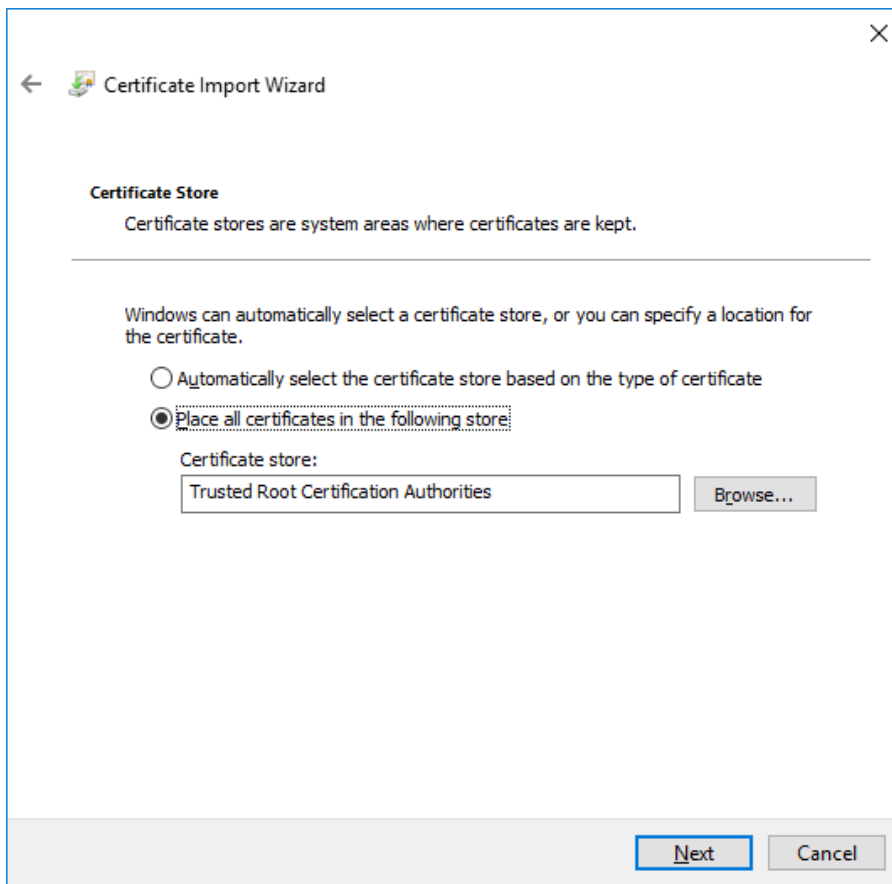
1. From the Certificates folder (Console Root > Certificates (Local Computer) > Trusted Root Certification Authorities), click Actions > All Tasks > Import.
2. The Certificate Import Wizard dialog appears.



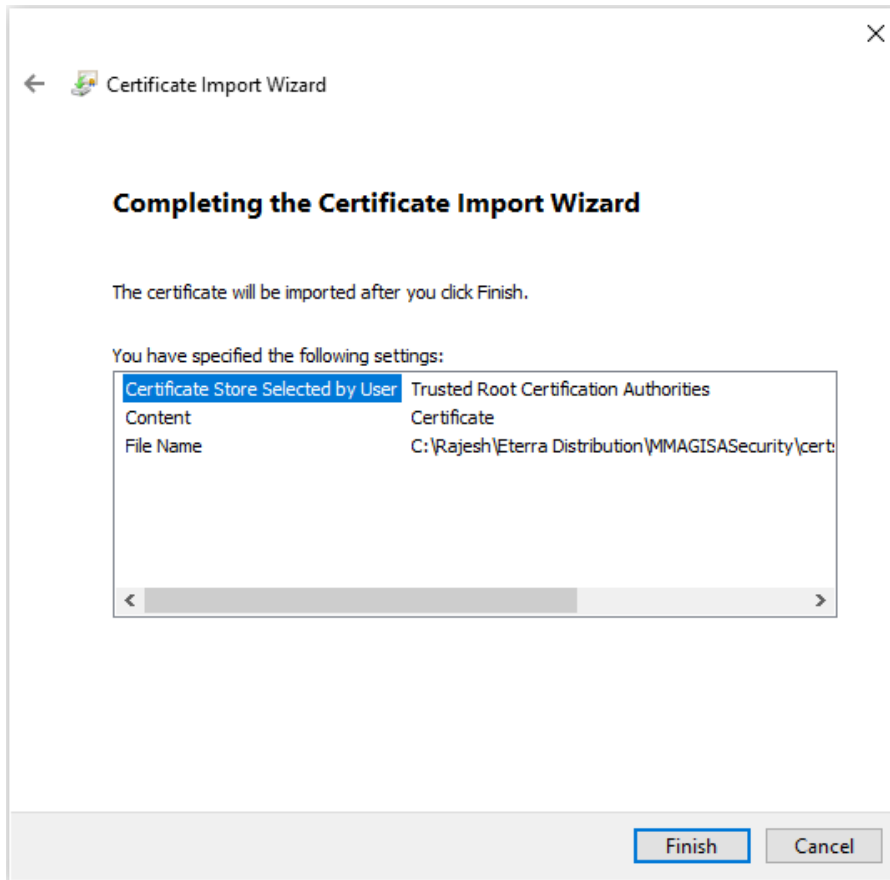
3. Click Next.
4. Click the Browse button and open the External System certificates (for example, `ca/BY0_Root_CA.crt`).
5. Click Next.



6. Select the "Place all certificates in the following store" option (selected by default).
7. Click Next.

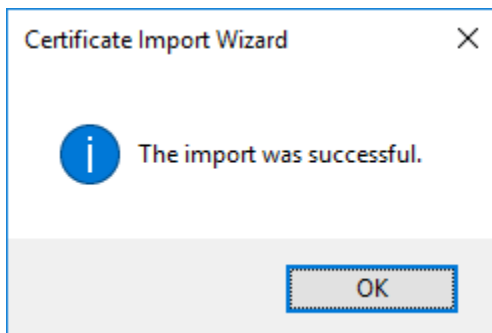


8. Click Finish.



The "The import was successful" pop-up appears.

9. Click OK.



10. Observe that the certificate is successfully imported to the Trusted Root Certification Authorities folder.
11. After importing, right-click the certificate, open the certificate properties, open the General tab, and change the Friendly Name parameter to ExternalSystemRESTInterfaceCA.

Console1 - [Console Root\Certificates (Local Computer)\Trusted Root Certification Authorities\Certificates]

File Action View Favorites Window Help

	Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name	Status
Console Root						
Certificates (Local Computer)						
Personal						
Certificates						
Trusted Root Certification Authorities						
Certificates						
Enterprise Trust						
Intermediate Certification Authorities						
Trusted Publishers						
Untrusted Certificates						
Third-Party Root Certification Authorities						
Trusted People						
	AAA Certificate Services	AAA Certificate Services	01-01-2029	Client Authenticati...	Sectigo (AAA)	
	AddTrust External CA Root	AddTrust External CA Root	30-05-2020	Client Authenticati...	Sectigo (AddTrust)	
	Assistant Root CA	Assistant Root CA	07-08-2023	<All>	<None>	
	Baltimore CyberTrust Root	Baltimore CyberTrust Root	13-05-2025	Client Authenticati...	DigiCert Baltimore Root	
	BYO_Root_CA	BYO_Root_CA	04-06-2022	<All>	ExternalSystemRESTInterfaceCA	
	BYO_Root_CA	BYO_Root_CA	05-04-2022	<All>	MMARESTInterfaceCA	
	Certum CA	Certum CA	11-06-2027	Client Authenticati...	Certum	
	Check MyTech SN	Check MyTech SN	26-04-2031	<All>	<None>	
	Class 3 Public Primary Certificat...	Class 3 Public Primary Certificatio...	02-08-2028	Client Authenticati...	VeriSign Class 3 Public Primary CA	
	COMODO RSA Certification Au...	COMODO RSA Certification Auth...	19-01-2038	Client Authenticati...	Sectigo (formerly Comodo CA)	

4.3. Sending REST messages to External Systems

Like MMA's REST server, the external system's REST server can be configured with CA-signed secure certificates. This allows the MMA REST client to verify the identity of the external system's REST server. The MMA REST client is also configured with an X.509 certificate that allows the server to verify the identity (client authentication) of the client. The external system's server certificate must be trusted by the MMA client and the MMA client's certificate must be trusted by the External System's server.

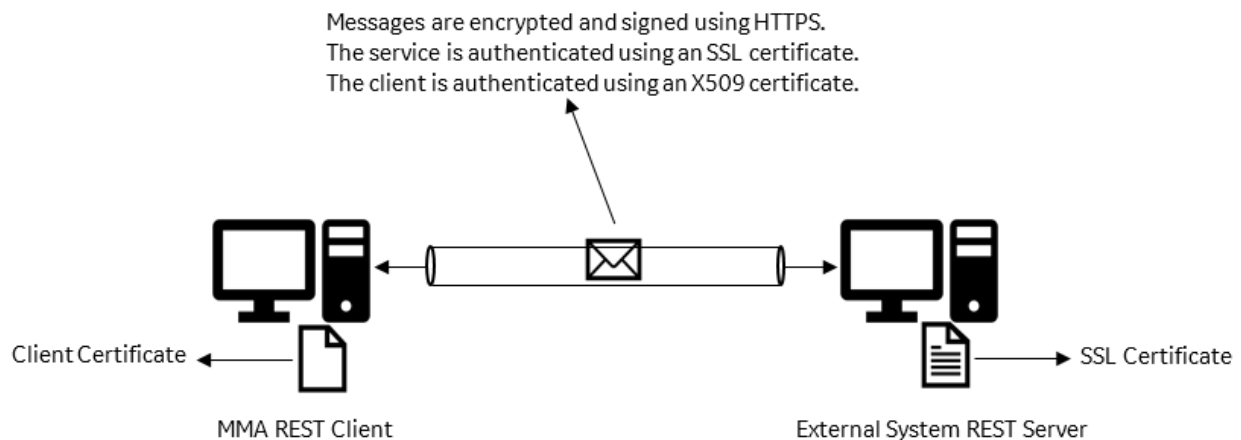


Figure 11. External System REST Server and MMA REST Client Interfacing using TLS with Certificate Authentication

Note

In the standard product configuration, the MMA uses only one SSL certificate for both binding with the service port and client authentication.

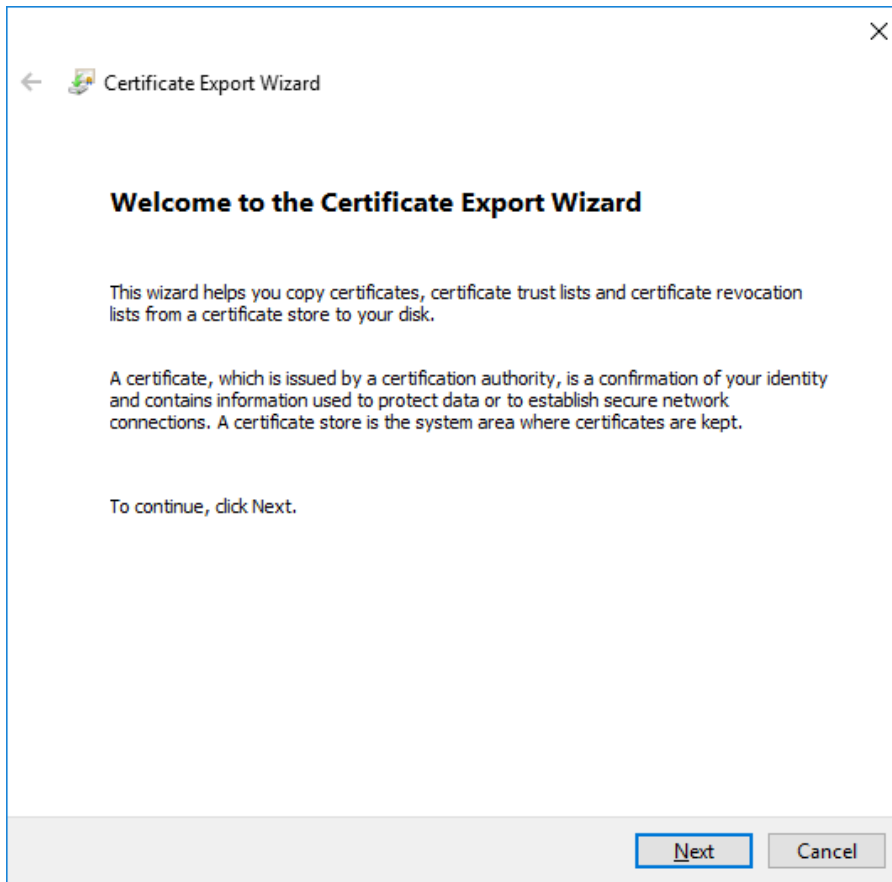
4.3.1. Exporting MMA Server Certificate for Client Authentication

During client authentication, the MMA sends SSL certificate to the external system.

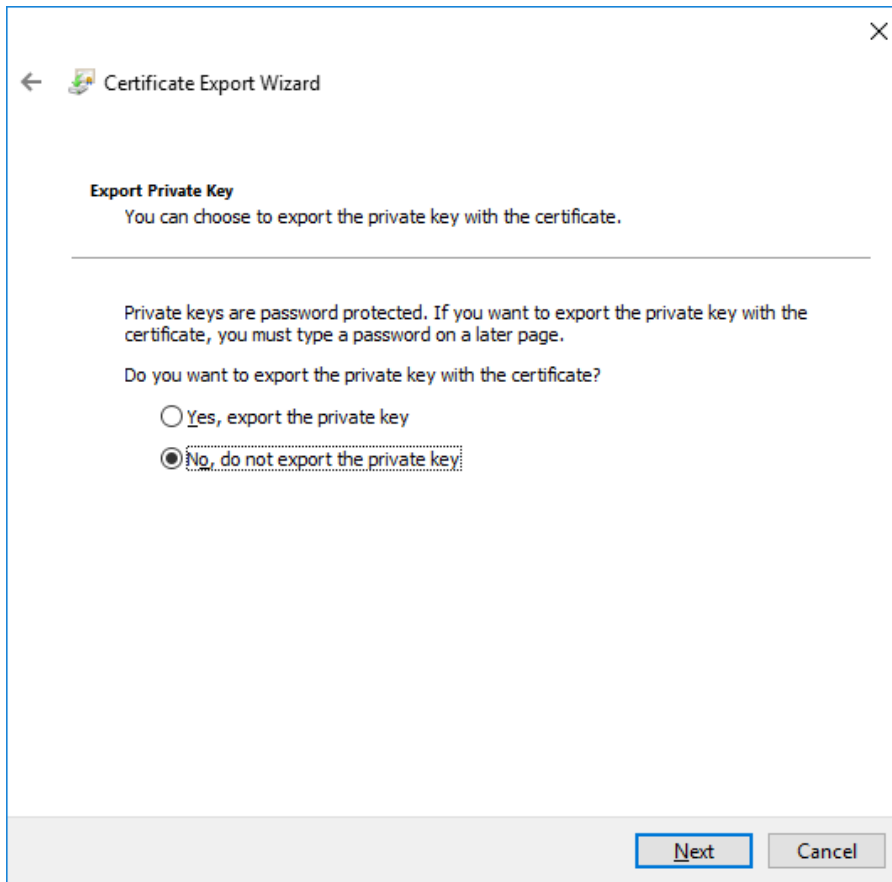
To configure a certificate:

1. Open the Microsoft Management Console application (for example, type mmc in the Windows Start menu).
2. In the Personal > Certificates folder, right-click MMARestInterface and click All Tasks > Export.

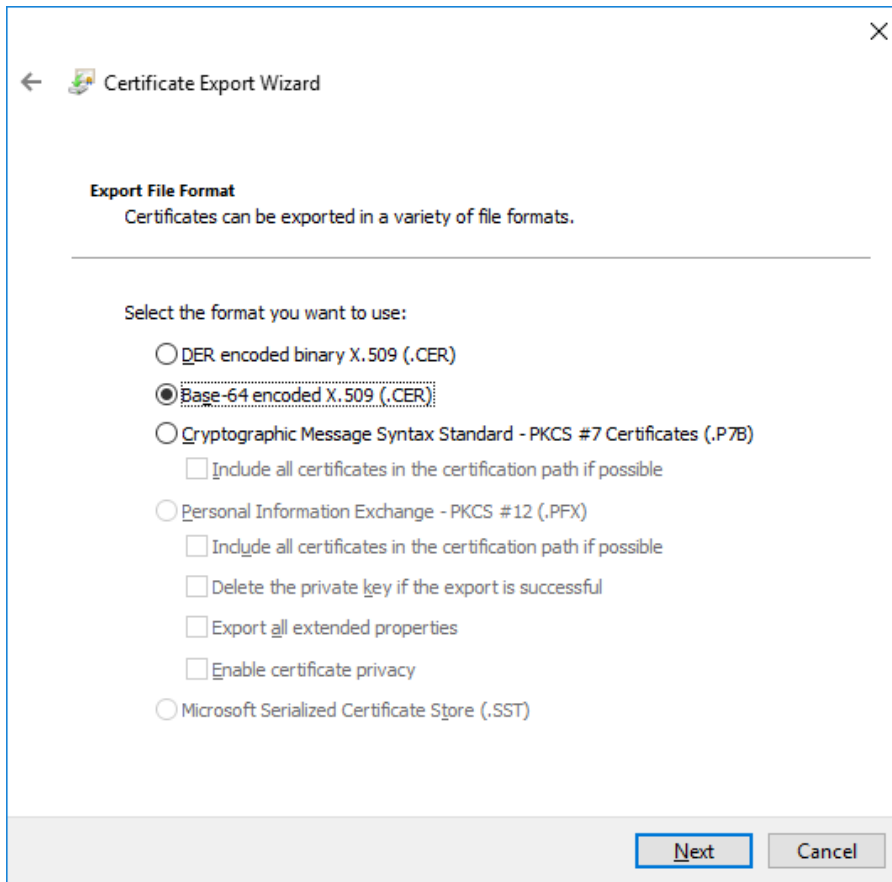
The Certificate Export Wizard appears.



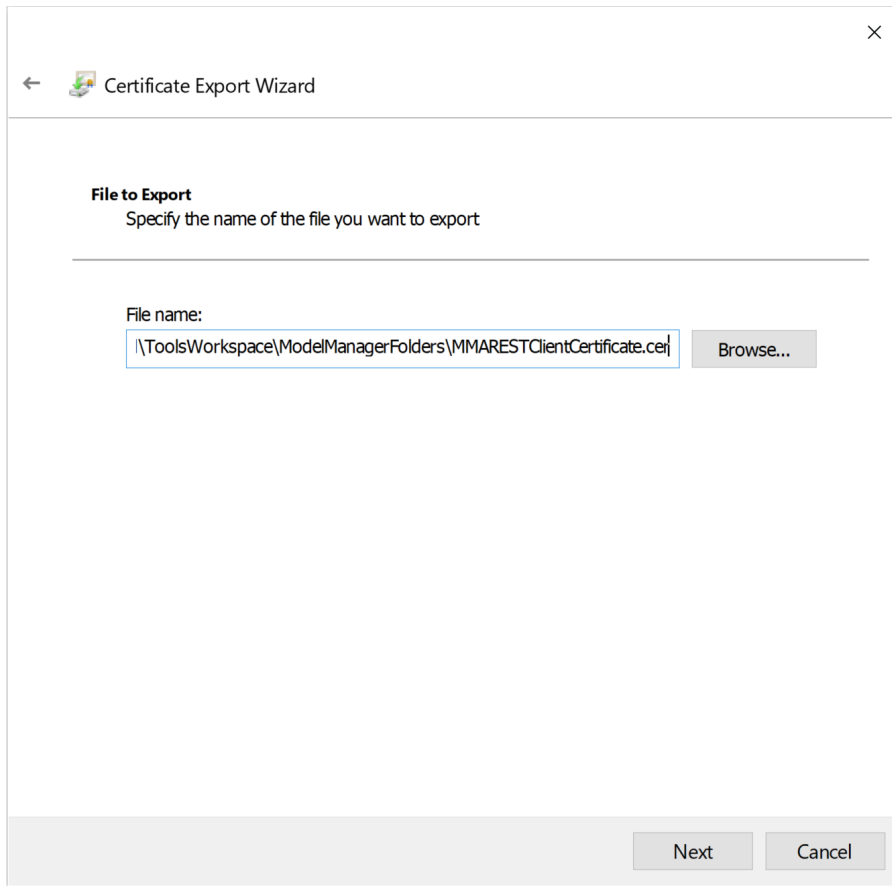
3. Click Next.
4. Select "No, do not export the private key".



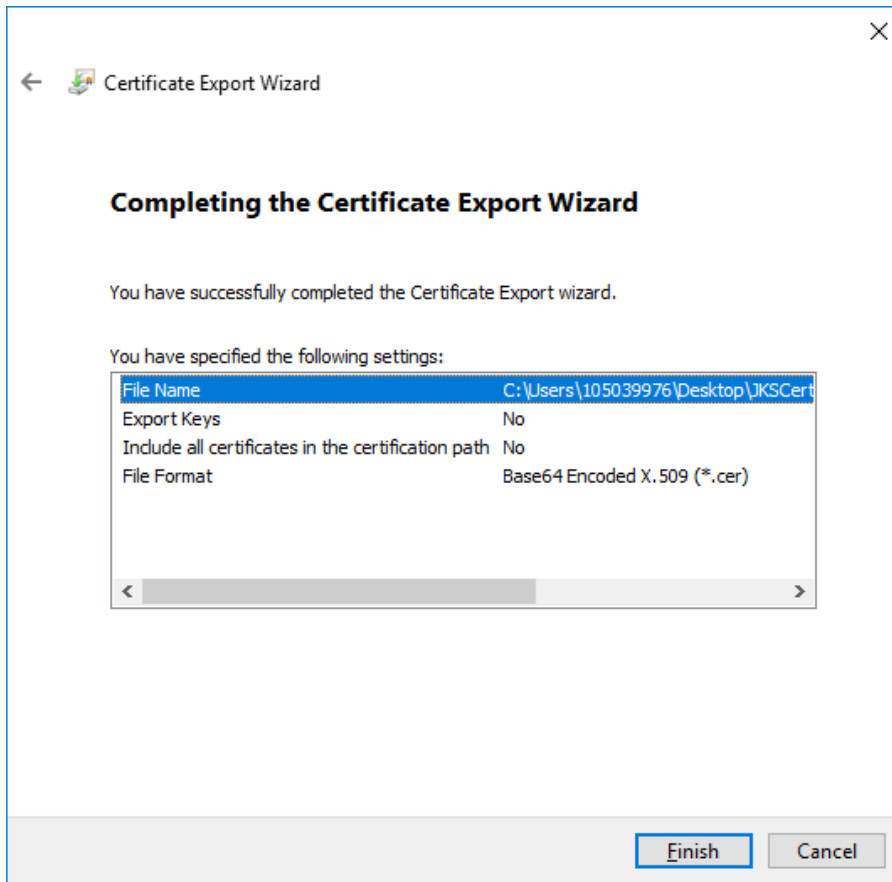
5. Click Next.
6. Select "Base-64 encoded X.509(.CER)".



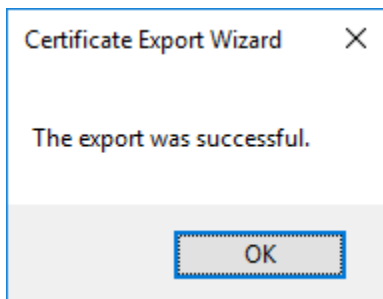
7. Click Next.
8. Click Browse, then browse to and select the certificate file.



9. Click Next.
10. Click Finish.



11. Click OK.



The certificate export process is completed. You can now use the exported file in the MMA Configuration > RESTful Interface > RESTful Interface Adapter > RESTful Client Certificate Details > Certificate File configuration in the Model Manager Configuration Editor.